Chapter 4 | Blanchard, Macroeconomics 9e

Session 1: Money Demand + Interest Rate Determination

Session 2: Banks, Reserves & Liquidity Traps

73-103 Principles of Macroeconomics | CMU-Q

Carnegie Mellon University Qatar

Financial Markets I

Why Money & Interest Rates Matter

Why Should You Care About Financial Markets?

The real-world impact on your daily life

- **Interest rates affect everything: mortgages, car loans, business investment**
- **Financial markets channel savings into productive investment**
- **Central bank decisions ripple through the entire economy**
- **Financial markets: bonds, stocks, currencies, mutual funds**

Example: Global Bond Market
~$130 Trillion
Larger than all stock markets

KEY INSIGHT Financial markets are the transmission mechanism between central bank policy and the real economy.

SECTION 4-1

The Demand for Money

Why do people hold money instead of interest-bearing assets?

- The money vs. bonds tradeoff
- Determinants of money demand
- The money demand function

Money vs. Bonds: The Fundamental Tradeoff

The simplifying assumption: only two financial assets

MONEY		BONDS
✓ **Yes: spend it now (liquid)**	**Use it for transactions?**	✗ No: must sell it first
✗ No: 0% return	**Pays interest?**	✓ **Yes: earns the interest rate *i***
✓ **None: value fixed in dollars**	**Risk of price change?**	✗ Yes: its price can change
Cash, checking accounts *Qatar: QAR cash, QNB checking*	**Examples**	T-bills, government bonds *Qatar: sovereign bonds*

Example: $10,000 for one year, *i* = 5%

As money: $10,000, and you can spend it any time

As a bond: $10,500 next year (+ $500 interest)

Opportunity cost of holding money = *i* × $10,000 = $500

LIQUIDITY convenience ⟷ **RETURN** interest

Higher *i* → convenience costs more → people hold less money (next slide).

KEY INSIGHT The choice between money and bonds depends on the tradeoff between liquidity (convenience) and return (interest).

What Drives Money Demand?

Two key determinants

1. Level of Transactions (Nominal Income, $Y)

- More income = more spending = need more money
- Money demand increases proportionally with $Y
- Think: Holiday shopping season vs. January

↑ $Y → ↑ Md

2. Interest Rate on Bonds (i)

- Higher i = higher opportunity cost of holding money
- People shift wealth from money to bonds
- Money demand decreases as i rises

↑ i → ↓ Md

The Money Demand Function

Formalizing the relationship

$$Md = \$Y \times L(i)$$

Money Demand = Nominal Income × Liquidity Function

Md
Money Demand
(how much money people want to hold)

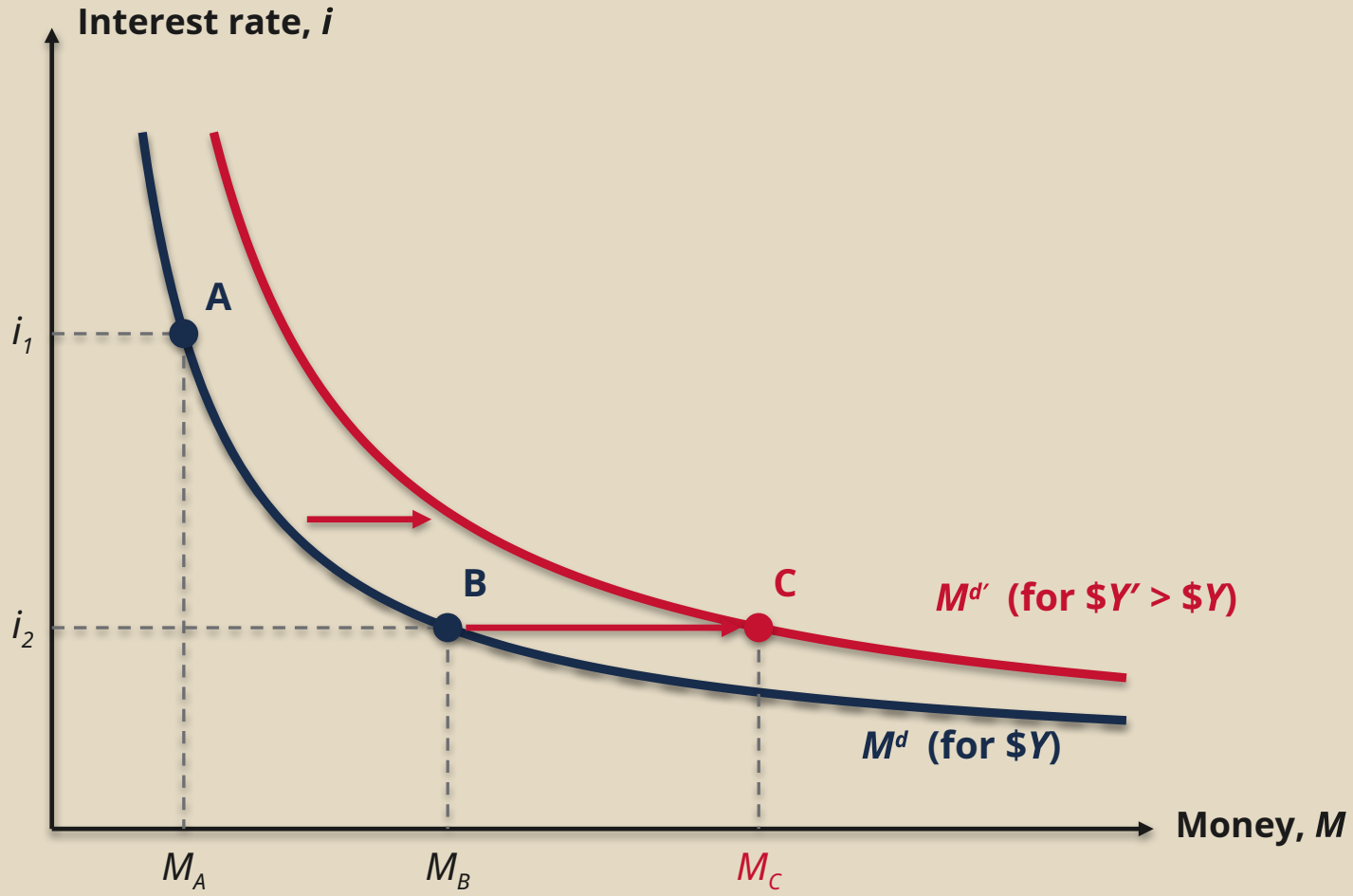
$Y
Nominal Income
(proportional: more income = more demand)

L(i)
Liquidity Function
(decreasing in i: higher i = less demand)

KEY INSIGHT L(i) is a decreasing function: as interest rates rise, people hold less money and more bonds.

Figure 4-1: The Demand for Money

Money demand as a function of the interest rate



Downward slope: A → B

Higher *i* = higher opportunity cost of holding money. As *i* falls, people hold more money.

Scenario: income rises ($Y ↑)

More income = more transactions = more money demanded at every interest rate.

Higher income shifts M^d right: at every interest rate, people want to hold more money.

Semantic Traps: Money ≠ Income ≠ Wealth

Getting the vocabulary right

Illustration: one household

Income
$5,000 per month

FLOW · measured per period

Saving
$1,000 per month

FLOW · per month

Spending
$4,000 per month

WEALTH = $200,000

MONEY = $3,000
cash + checking

Everything you own, measured today

Other assets = $197,000
bonds, stocks, house, …

STOCK · measured on a date

Units test

A flow has a time unit (per month, per year). A stock is measured on a date (e.g., 31 December).

Trap 1

"She makes a lot of money." → She has a high INCOME (a flow).

Trap 2

"He has a lot of money." → He has a lot of WEALTH (a stock).

In this course

Money = cash + checking deposits: the assets you can use for transactions right away.

KEY INSIGHT

Money is a STOCK (what you hold now); income is a FLOW. Money demand = how much of your wealth you choose to hold as money.

Think-Pair-Share: What Happens to Money Demand?

ACTIVITY

1. Qatar's nominal GDP doubles due to a new LNG expansion. What happens to Md?

2. The QCB raises interest rates from 5% to 7%. What happens to Md?

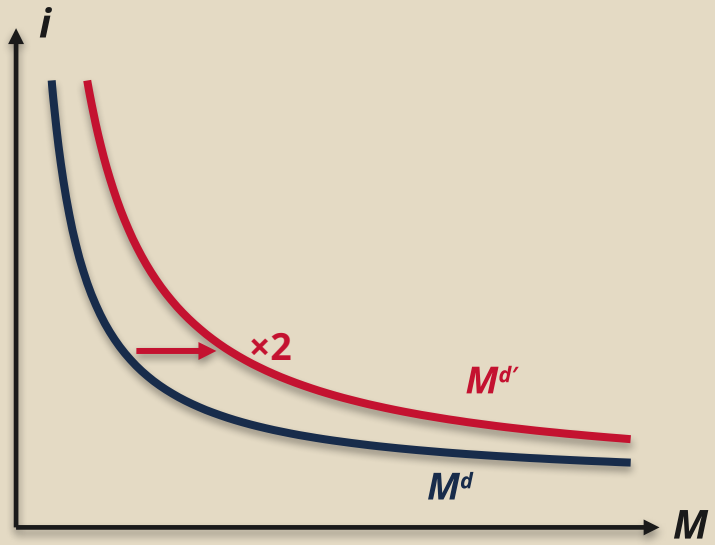
3. A new mobile payment app makes it easier to convert bonds to cash instantly. What happens to L(i)?

Discuss in pairs (3 minutes). Be ready to explain using the Md = $Y × L(i) framework.

Think-Pair-Share: Answers

Shift the curve, or move along it?

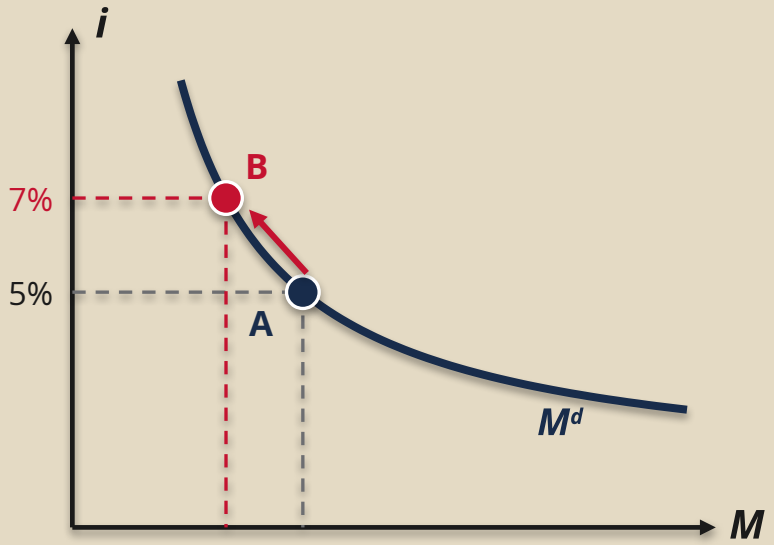
Q1 · Nominal GDP doubles (LNG expansion)



M^d shifts RIGHT

$M^d = \$Y \times L(i)$: if $\$Y$ doubles, money demand doubles at every interest rate.

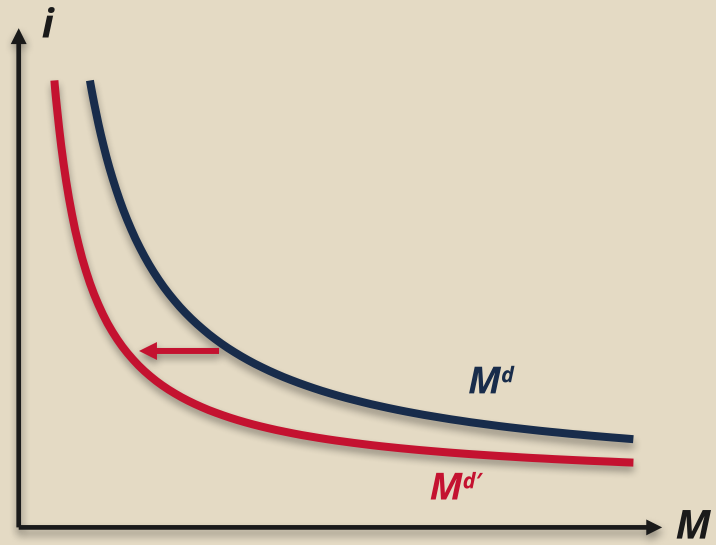
Q2 · QCB raises *i* from 5% to 7%



Movement ALONG M^d (A → B)

Higher *i* = higher opportunity cost of holding money, so people hold less. The curve itself does not move.

Q3 · Payment app: bonds → cash instantly



M^d shifts LEFT: $L(i)$ falls

If bonds turn into cash instantly, people need to hold less money at every interest rate.

$Y changes, or a new payment technology → the M^d curve SHIFTS. *i* changes → we MOVE ALONG it.

Quick Check: Money Demand

Test your understanding

True or False: Money demand rises with income.

?

True or False: Higher interest rates increase money demand.

?

If $\$Y = \$1{,}000$ and $L(i) = 0.5$, what is M^d?

?

Commit to an answer for each one before we reveal.

Quick Check: Money Demand

Test your understanding

True or False: Money demand rises with income.	**TRUE**	More income → more transactions → more money needed. M^d is proportional to $\$Y$.
True or False: Higher interest rates increase money demand.	**FALSE**	Bonds pay i, money pays 0. Higher i = higher opportunity cost, so people hold less money: $L(i)$ falls as i rises.
If $\$Y$ = \$1,000 and $L(i)$ = 0.5, what is M^d?	**\$500**	M^d = \$1,000 × 0.5 = \$500. $L(i)$ = 0.5 means people hold money worth half of their nominal income.

Follow-up 1: i rises so that $L(i) = 0.4$ → M^d = \$1,000 × 0.4 = \$400 (a movement along M^d)

Follow-up 2: $\$Y$ doubles to \$2,000 → M^d = \$2,000 × 0.5 = \$1,000 (the M^d curve shifts right)

KEY INSIGHT Money demand: proportional to \$Y, decreasing in i. The formula Md = \$Y × L(i) captures both relationships.

From Demand → Equilibrium

We know what determines money demand.
Now: How does the interest rate get determined?

Answer: Where money supply meets money demand.

SECTION 4-2

Determining the Interest Rate I

The simplest model: money supply, money demand, and equilibrium

- Money market equilibrium
- Effects of income and money supply changes
- Open market operations

Equilibrium in the Money Market

Where supply meets demand

$$Ms = Md \quad \rightarrow \quad Ms = \$Y \times L(i)$$

Money Supply (set by central bank) = Money Demand

Money Supply (Ms)

Set by the central bank as a policy decision. Vertical line on graph.

Money Demand (Md)

Determined by income ($Y) and interest rate (i). Downward-sloping curve.

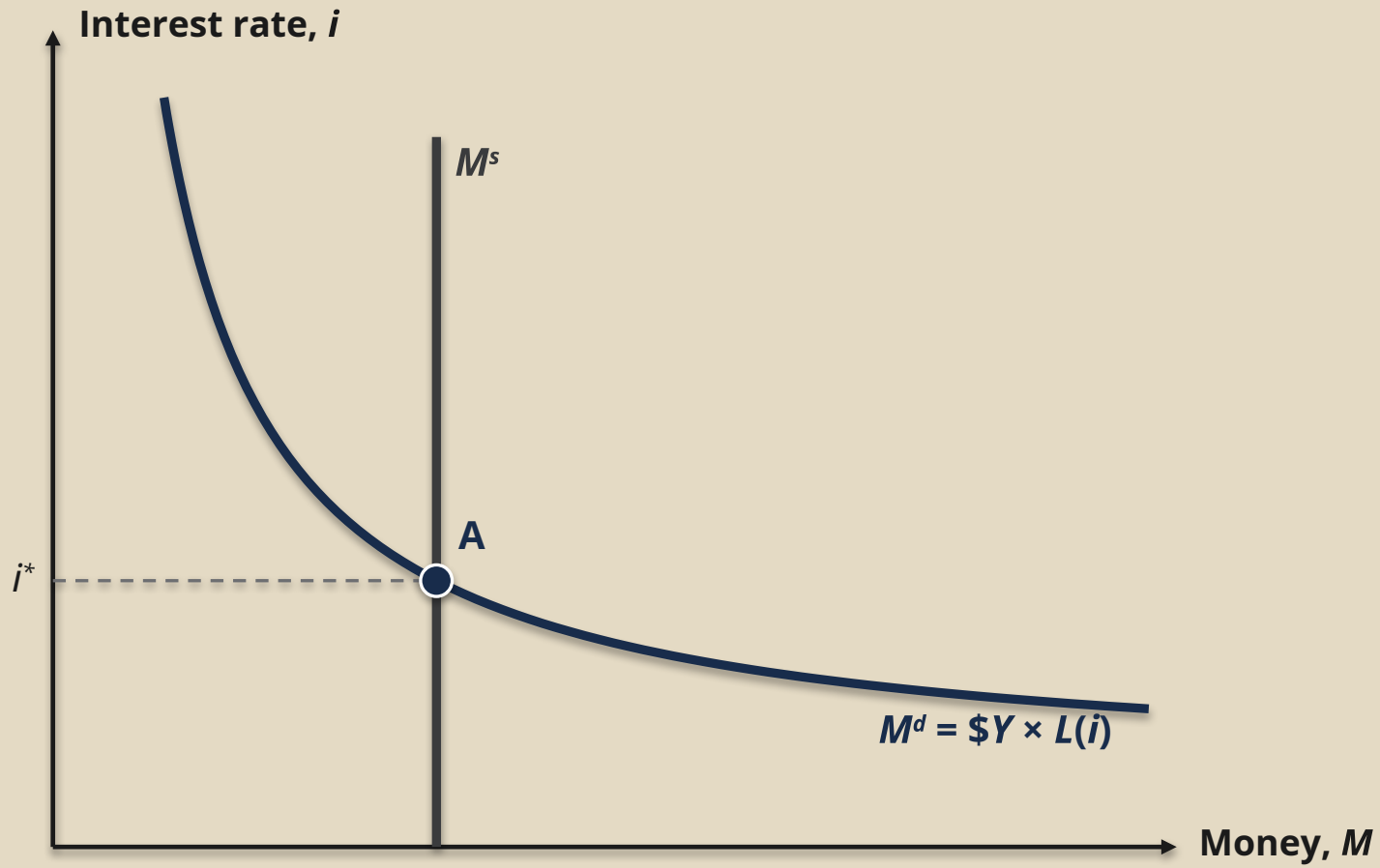
Equilibrium (i*)

The interest rate where Ms = Md. Found at intersection.

KEY INSIGHT The central bank controls Ms. Given $Y, the equilibrium interest rate i* adjusts so people willingly hold exactly Ms.

Figure 4-2: Determination of the Interest Rate

Money market equilibrium



M^d slopes down

Higher i = higher opportunity cost, so people hold less money.

M^s is vertical

The central bank fixes the money supply (a policy decision). It does not depend on i.

Equilibrium A: $M^s = M^d$

The intersection pins down the interest rate i^*.

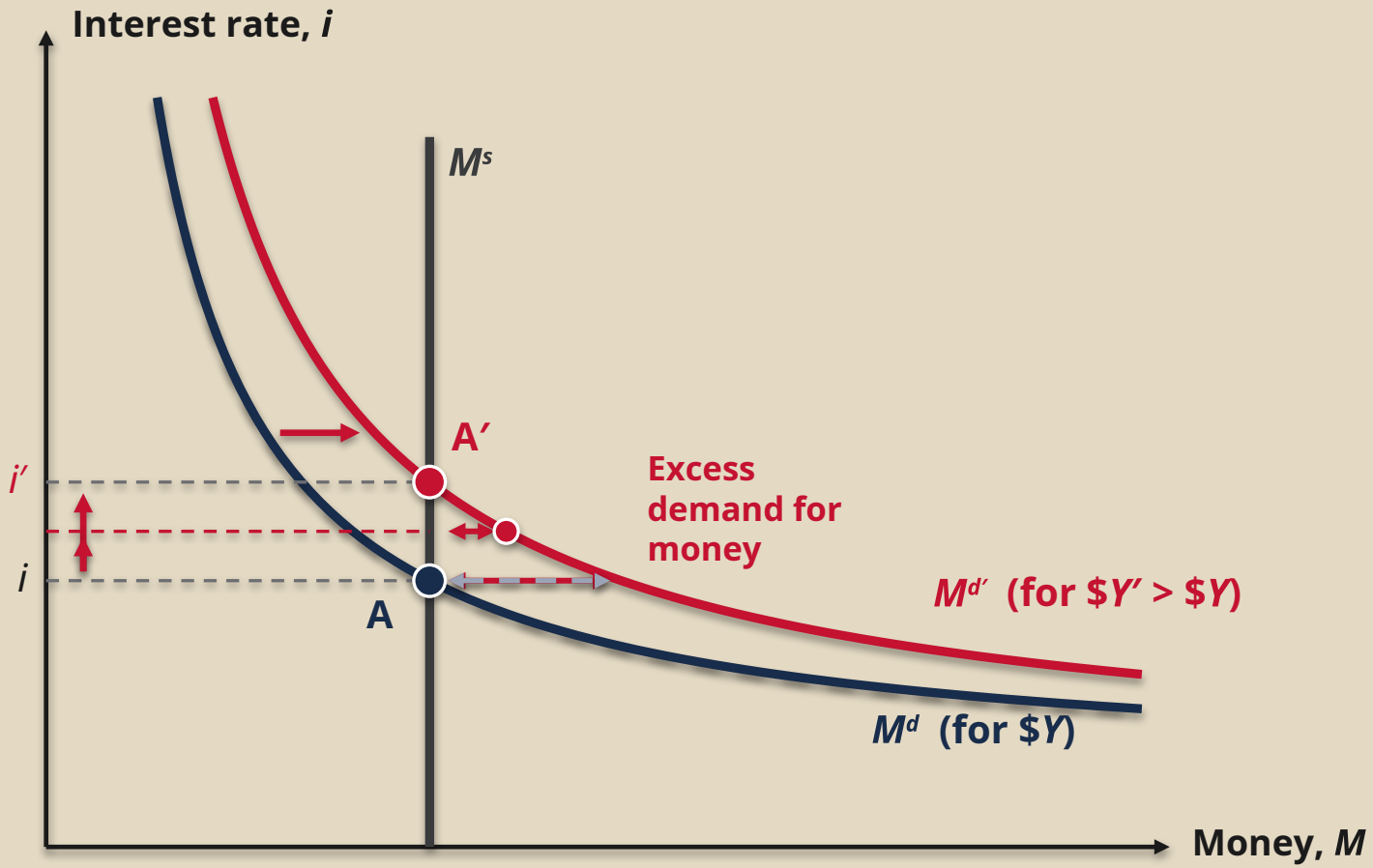
The central bank controls M^s. Given $\$Y$, i^* adjusts so people willingly hold exactly M^s.

Figure 4-3: What Happens When Income Rises?

Effect of an increase in nominal income on the interest rate

Figure 4-3: What Happens When Income Rises?

Effect of an increase in nominal income on the interest rate



Shock: income rises ($*Y* ↑)

More transactions → more money demanded at every *i*.

Adjustment through the bond market

Excess demand for money → SELL bonds

$$i = \frac{\$100 - P_B}{P_B}$$

P_B ↓

⇒ *i* ↑

P_B = price of the bond today
\$100 = what the bond pays in one year
i = the bond's yield = our interest rate

e.g. P_B falls \$96 → \$95, so *i* rises 4.2% → 5.3%

Higher *i* → holding money costs more → quantity demanded falls along $M^{d'}$.

Selling stops at A′: $M^{d'} = M^s$ at the higher i'. People willingly hold the existing money.

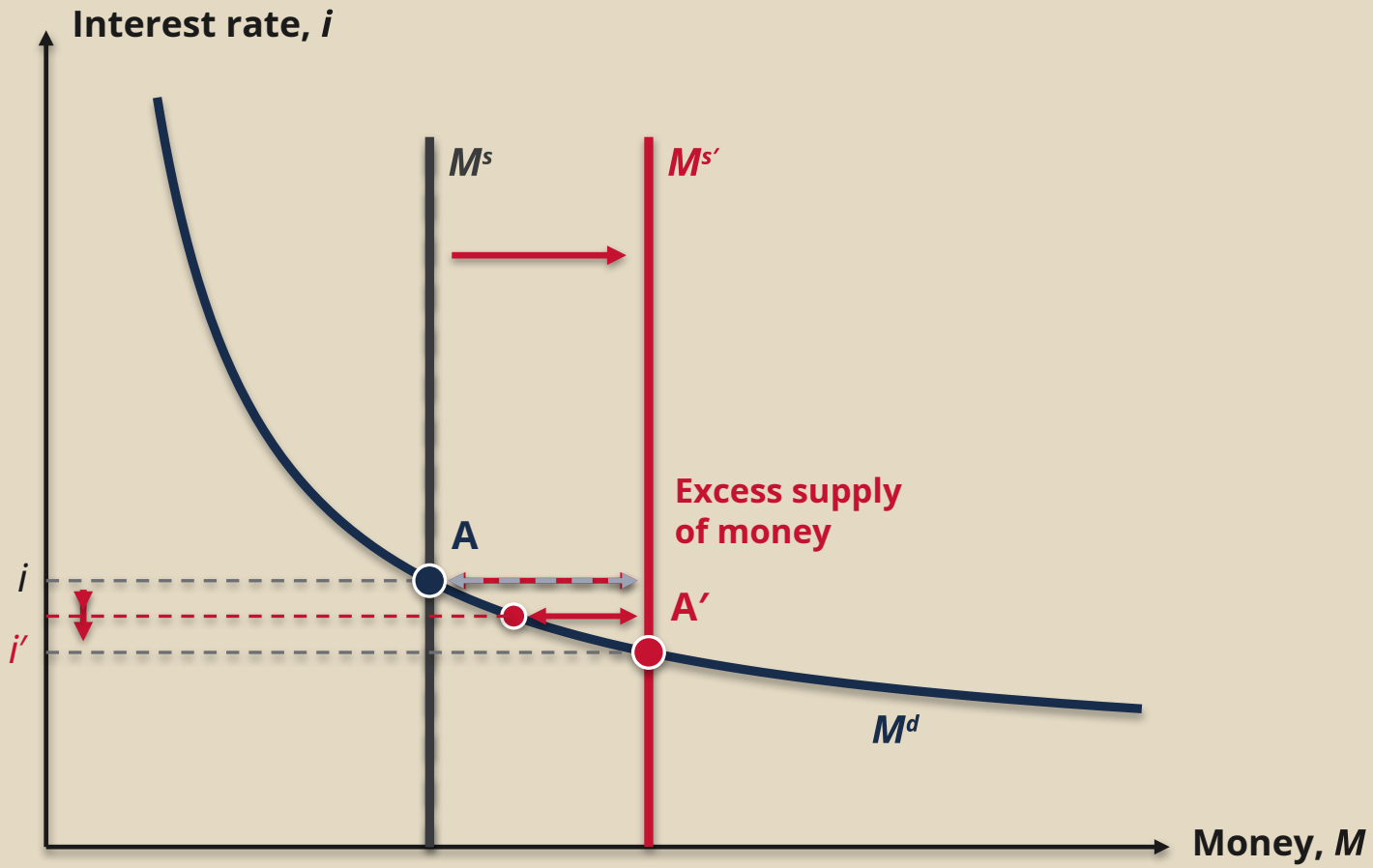
Real world: an economic boom pushes interest rates up, unless the central bank expands M^s.

Figure 4-4: What Happens When Money Supply Rises?

Effect of an increase in money supply on the interest rate

Figure 4-4: What Happens When Money Supply Rises?

Effect of an increase in money supply on the interest rate



Policy: central bank raises M^s

Money supply shifts right; money demand is unchanged.

Adjustment through the bond market

Excess supply of money → BUY bonds

$$i = \frac{\$100 - P_B}{P_B}$$

P_B ↑ ⇒ i ↓

P_B = price of the bond today
\$100 = what the bond pays in one year
i = the bond's yield = our interest rate

e.g. P_B rises \$95 → \$96, so i falls 5.3% → 4.2%

Lower i → holding money costs less → quantity demanded rises along M^d.

Buying stops at A′: $M^d = M^{s'}$ at the lower i'. People willingly hold the new money.

This is how central banks stimulate the economy!

Open Market Operations

How central banks change the money supply

The central bank's balance sheet

Assets	Liabilities
Bonds	Currency (money)
+ $1 million bonds bought	**+ $1 million** new currency

① The central bank buys $1 million of bonds from the public.

② It pays with money it creates: M^s rises by $1 million.

The public now holds more money and fewer bonds

↓

Demand for bonds ↑ → bond prices ↑

↓

Interest rate i ↓
why? next slide: $P_B = \$100 / (1 + i)$

Expansionary OMO
CB buys bonds → M^s ↑ → bond prices ↑ → i ↓

Contractionary OMO (mirror image)
CB sells bonds → M^s ↓ → bond prices ↓ → i ↑

Open market operations are how the central bank shifts M^s (Figure 4-4): buy bonds → M^s ↑ → i ↓.

Bond Prices and Interest Rates

The inverse relationship

$$P_B = \frac{\$100}{1 + i}$$

A one-year bond (T-bill) that pays $100 in one year

P_B price of the bond today

$100 face value, paid in one year

i the one-year interest rate

Same equation as in Figures 4-3 and 4-4

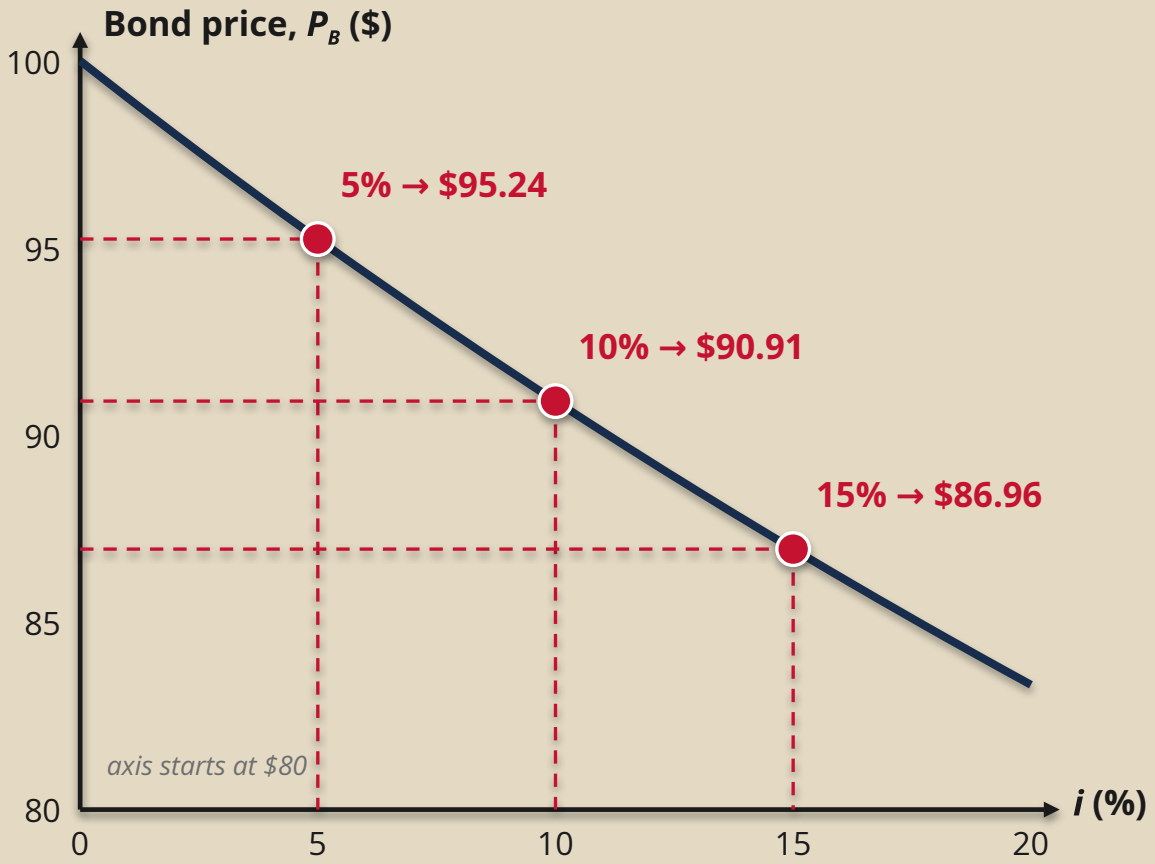
Pay P_B today, get $100 in a year.

Your interest rate is:

$$i = (\$100 - P_B) / P_B.$$

Solve for the price:

$$P_B = \$100 / (1 + i).$$



Bond price ↑ ⇔ interest rate ↓. When the central bank buys bonds, P_B rises, so i falls.

Focus: Who Holds US Currency?

Puzzle: $2.39 trillion of US cash exists. Who is holding it?

US currency in circulation

$2.39 trillion 31 Dec 2025

÷ 341.8 million people (July 2025)

≈ $7,000 in cash per person

Do you hold $7,000 in cash? Does a family of four hold $28,000? So who holds it?

Clue: 83% of that value is in $100 notes.

Sources: Federal Reserve Board, Currency in Circulation (31 Dec 2025); US Census Bureau, Vintage 2025 estimates; Federal Reserve, "The International Role of the U.S. Dollar – 2025 Edition" (FEDS Notes, Jul 2025); Fed IFDP 1387 (2024).

Where is US cash held? (Fed staff estimate, early 2025)



Estimates are uncertain: Fed researchers report a range of about 40–75%.

Why abroad?

Dollars are a safe store of value where local currencies are unstable.

Link to M^d

Foreign holdings grew more slowly as US rates rose: a higher *i* raises the opportunity cost of holding cash (Fed, 2025).

Who gains?

Cash pays no interest, so foreigners holding dollars give the US an interest-free loan (seigniorage).

KEY INSIGHT The global demand for US dollars gives the Fed extraordinary influence over world financial markets.

Case Study: Central Bank Rate Decisions

CASE STUDY

Fed rate hikes, 2022–23

To fight high inflation, the Fed raised its target range for the federal funds rate 11 times: from 0.25–0.50% (March 2022) to 5.25–5.50% (July 2023).

Q: Using our model, what did the Fed do to M^s? Draw the graph.

Discuss in groups (5 minutes). Use the M^s / M^d framework.

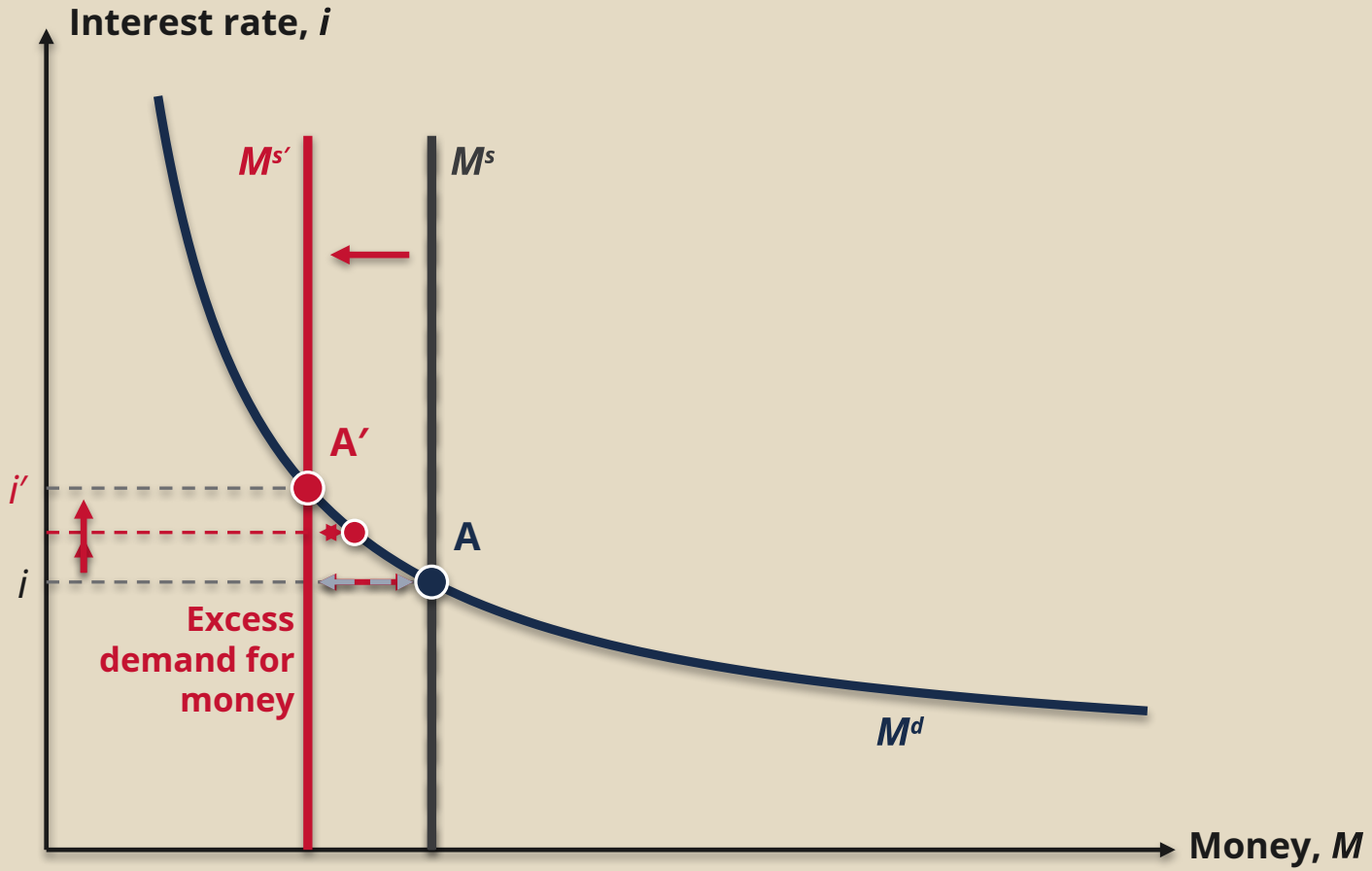
Fed funds target range, upper bound (%)



Source: Federal Reserve Board, FOMC target range history.

Case Study Solution: The Fed Tightens

Contractionary monetary policy in the money market



Example: 1-year Treasury yield, 3 Jan 2022 and 27 Jul 2023 (FRED, DGS1); price implied by $P_B = \$100/(1 + i)$.

Policy: the Fed lowers M^s

It sells bonds (contractionary OMO); money demand is unchanged.

Adjustment through the bond market

Excess demand for money → SELL bonds

$$i = \frac{\$100 - P_B}{P_B} \qquad P_B \downarrow \Rightarrow i \uparrow$$

P_B = price of the bond today
\$100 = what the bond pays in one year
i = the bond's yield = our interest rate

T-bill: P_B \$99.60 → \$94.88, so i 0.4% → 5.4%

Higher i → holding money costs more → quantity demanded falls along M^d.

Selling stops at A′: $M^d = M^{s'}$ at the higher i'. People hold the smaller money stock.

In practice the Fed announces a target range and adjusts its tools to hit it (Section 4-3). In our model, that is a leftward shift of M^s.

What Is a Bond Yield?

The return you earn on a bond

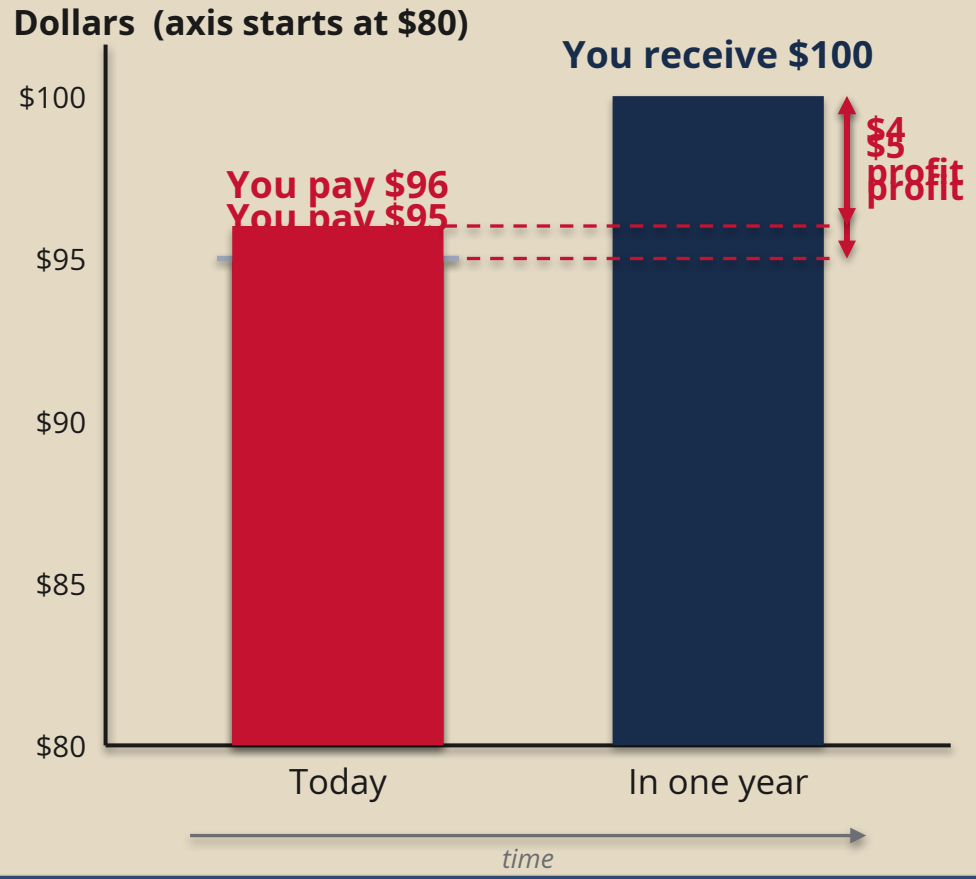
Yield = the return you earn on a bond
Plain English: your profit ÷ what you paid.

From the bond-price equation (slide 22) to the yield

1. Start: $P_B = \$100 / (1 + i)$
2. Rearrange: $1 + i = \$100 / P_B$
3. Solve: $i = (\$100 - P_B) / P_B$
4. **Pay $95: $i = \$5 / \$95 = 0.053 = 5.3\%$**

Same equation as on slides 18 and 20 (the money-market adjustment).

Think about it: if new bonds pay 7%, would anyone still pay $95 for a bond that yields 5.3%?



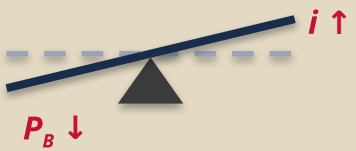
Pay $96: yield = $4 / $96 = 4.2% (price ↑ → yield ↓)

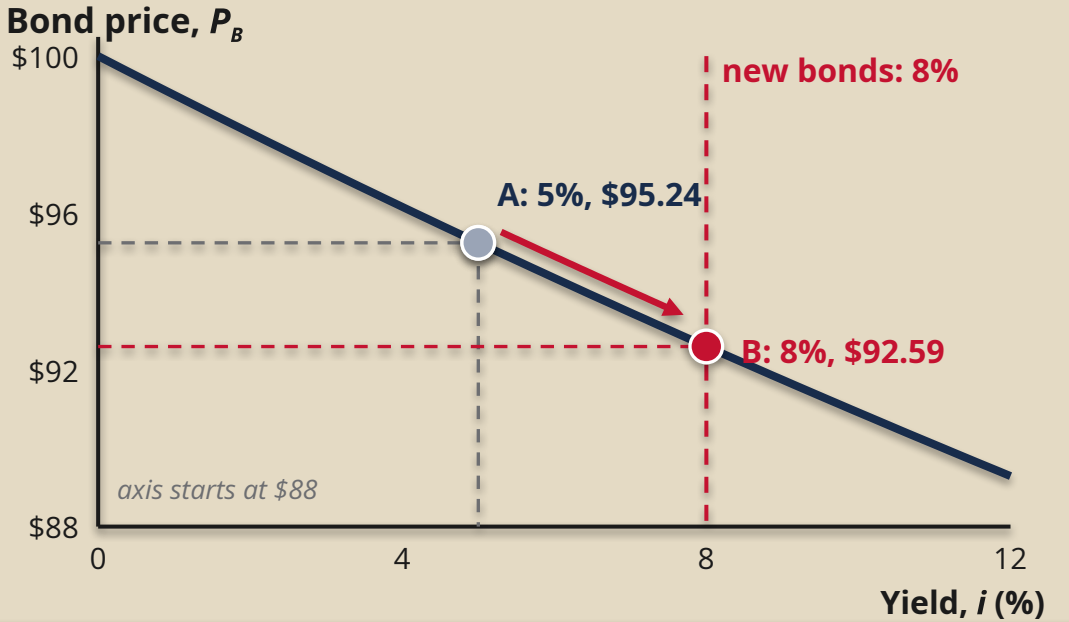
Yield = (what you get − what you pay) ÷ what you pay. Pay more for the same $100 → lower yield.

The Inverse Relationship: Bond Prices vs. Yields

Why they always move in opposite directions

1. Old bond: pays $100 in one year. Price $95.24 → yield 5%.
2. The central bank raises rates: NEW bonds now yield 8%.
3. Who pays $95.24 to earn 5% when new bonds earn 8%? Nobody. Sellers must cut the price.
4. The price falls to $92.59. Now the old bond also yields 8%: $100 / 1.08 = $92.59.
5. **The $100 never changes; only the price adjusts. Price ↓ ⇔ yield ↑.**



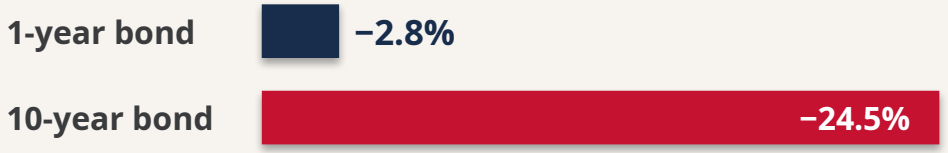


Why it matters: Silicon Valley Bank, March 2023

SVB had put depositors' money into long-term bonds bought when yields were low. As rates rose, those bonds lost value. Depositors rushed to withdraw, and the bank failed.

Same jump, 5% → 8%, for a 10-year bond

Pays $100 in 10 years: $P_B = \$100 / (1 + i)^{10}$ → $61.39 → $46.32

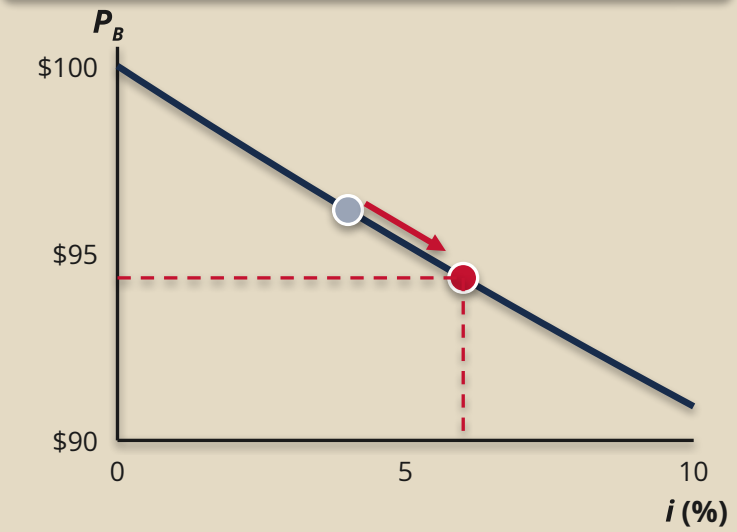


Rates up → existing bond prices down. The longer the bond, the bigger the fall.

Bond Prices & Yields in the News

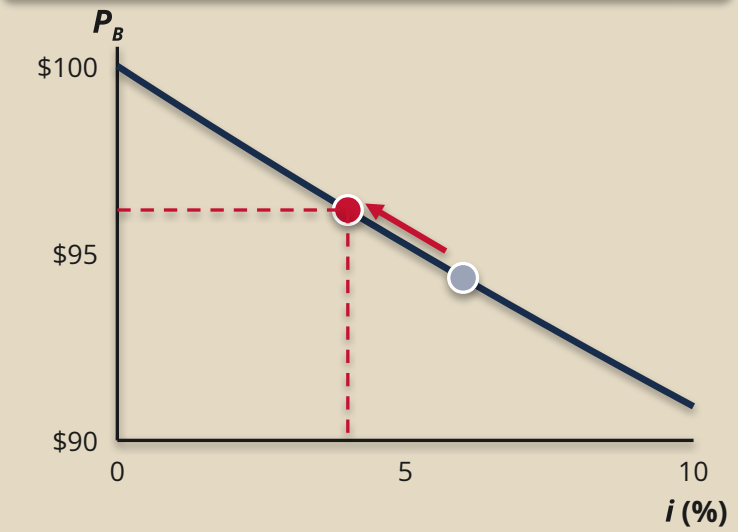
Read the headline, then move along the curve

"Treasury yields jump from 4% to 6%"



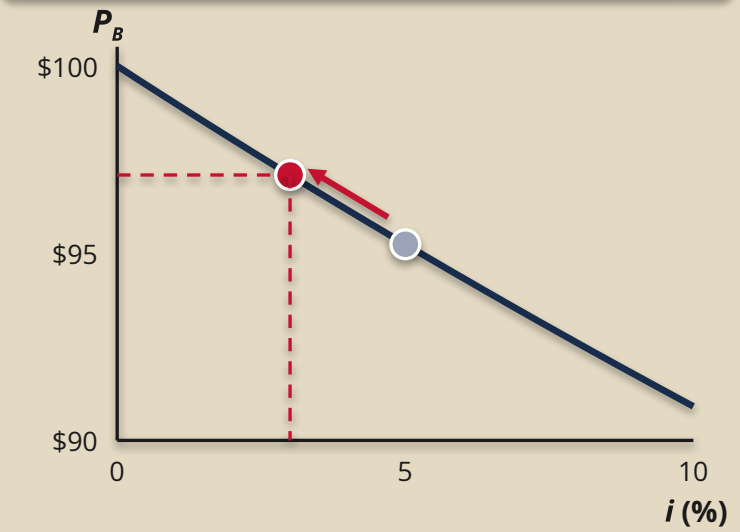
Yield ↑ ⇒ price ↓
$96.15 → $94.34. Bond prices FELL.

"Bond rally: prices climb from $94.34 to $96.15"



Price ↑ ⇒ yield ↓
6% → 4%. Bond yields FELL.

"The Fed cuts rates: new bonds now yield 3%, not 5%"



Old 5% bonds are worth more
Priced to yield 3%: $95.24 → $97.09.

Quick reference, P_B = $100 / (1 + i): 2% → $98.04 4% → $96.15 5% → $95.24 6% → $94.34 8% → $92.59 10% → $90.91

Yields and prices are two ways of reading the same bond: when one rises, the other falls.

What Is the Yield Curve?

Yields across bond maturities

The yield curve plots the yields on government bonds of different maturities, from 3 months to 30 years, on one day.

Why do longer bonds usually have HIGHER yields?

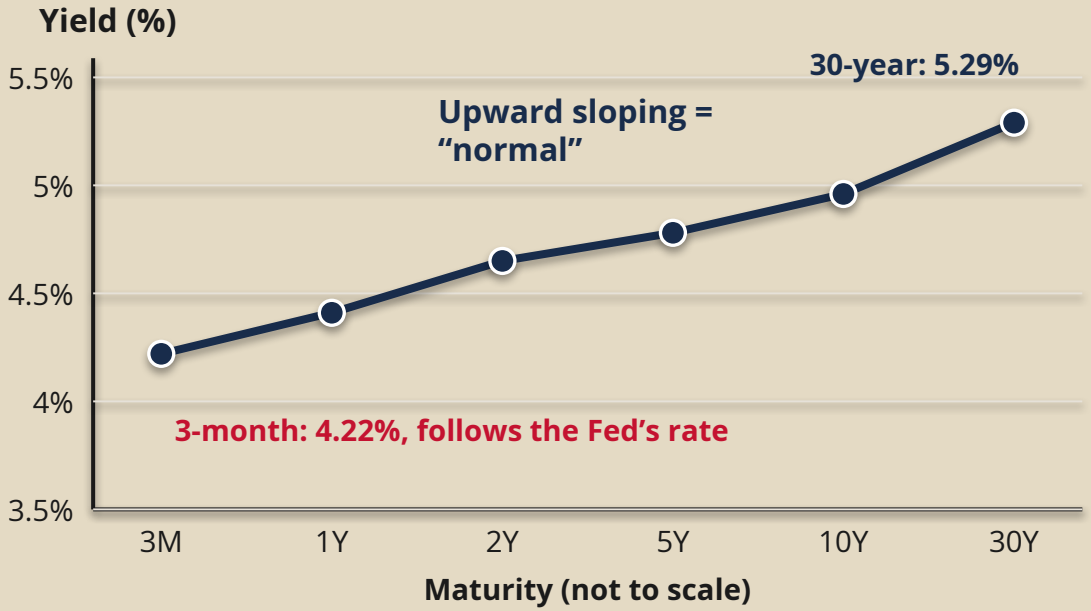
1. Expectations

A long rate ≈ the average of the short rates expected over its life. Today 4%, expected 5% next year → 2-year yield ≈ 4.5%.

2. Risk (term) premium

Long bonds are riskier to hold: their prices swing more when rates move (slide 27: −24.5% vs. −2.8%), and inflation is harder to predict far ahead. Investors want a little extra.

Usually both push the curve up. But if markets expect the Fed to CUT rates, reason 1 pulls long yields BELOW short ones → the curve inverts (next slides).



US Treasury yields, 22 Oct 2026 (FRED). Axis starts at 3.5%.

Blanchard 14-2: the 2-year yield



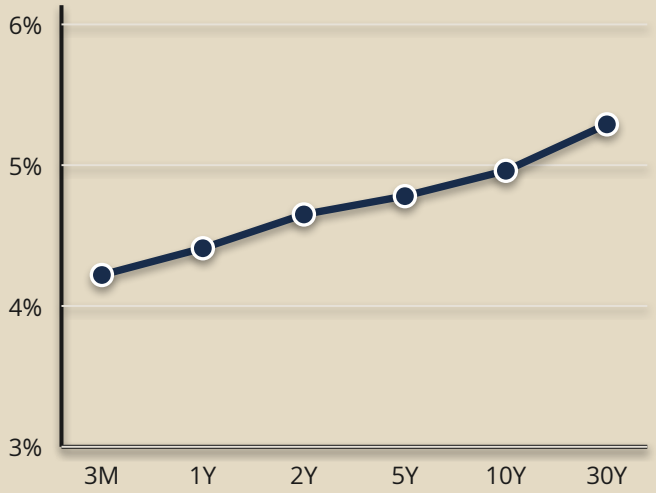
Short end: the Fed's rate today. Long end: what markets expect the Fed to do later, plus a premium.

The Three Shapes of the Yield Curve

Real US Treasury curves. What does each one signal?

Normal (upward)

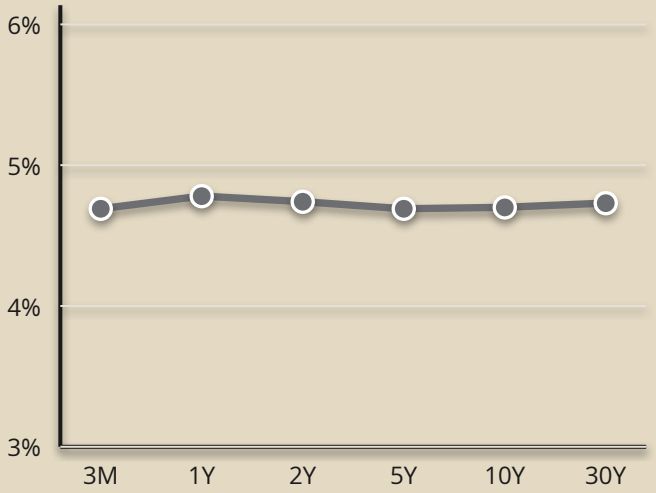
22 Oct 2026



Signal?

Flat

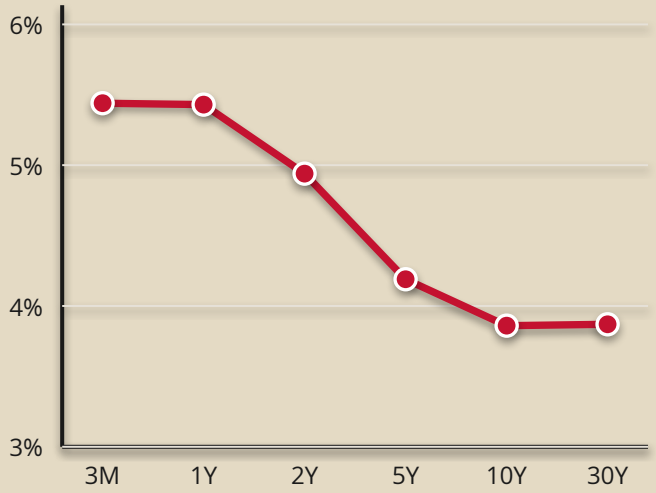
22 Mar 2006



Signal?

Inverted (downward)

3 Jul 2023



Signal?

Pairs (2 minutes): for each shape, where do markets expect the Fed's rate to go?

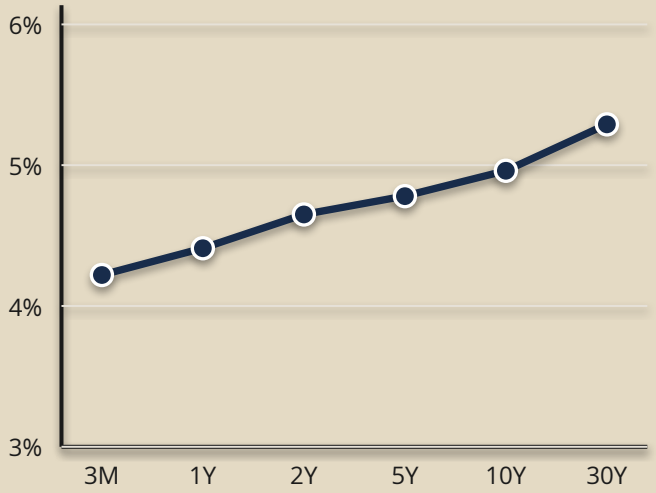
Source: FRED, US Treasury constant-maturity yields (DGS3MO … DGS30). Yield axis 3–6%, maturities not to scale.

The Three Shapes of the Yield Curve

Each shape is a forecast of the Fed's future rate

Normal (upward)

22 Oct 2026

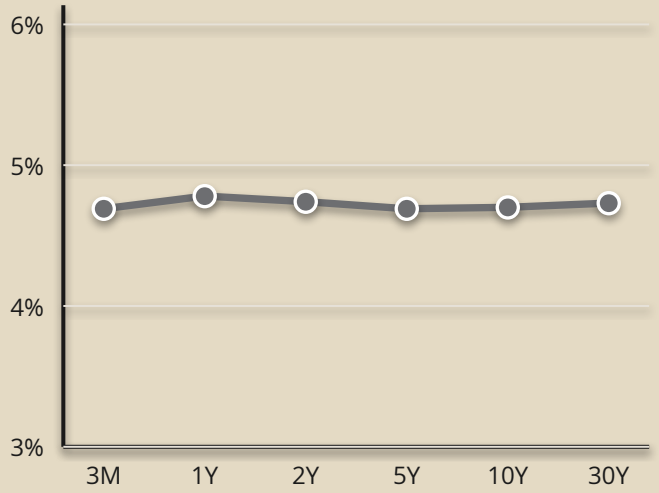


Expected Fed rate: rise or stay

Normal times
Short rates expected to rise or stay, plus a premium → long > short.

Flat

22 Mar 2006

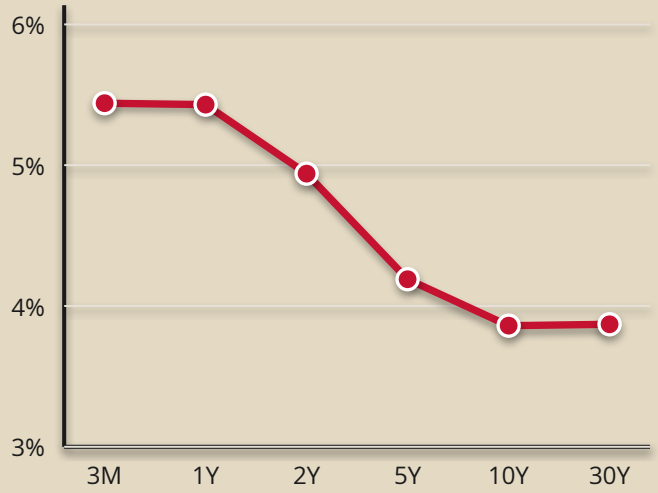


Expected Fed rate: stay put

A turning point
Short rates expected to stay put. Mar 2006: near the end of the 2004–06 hikes.

Inverted (downward)

3 Jul 2023



Expected Fed rate: fall

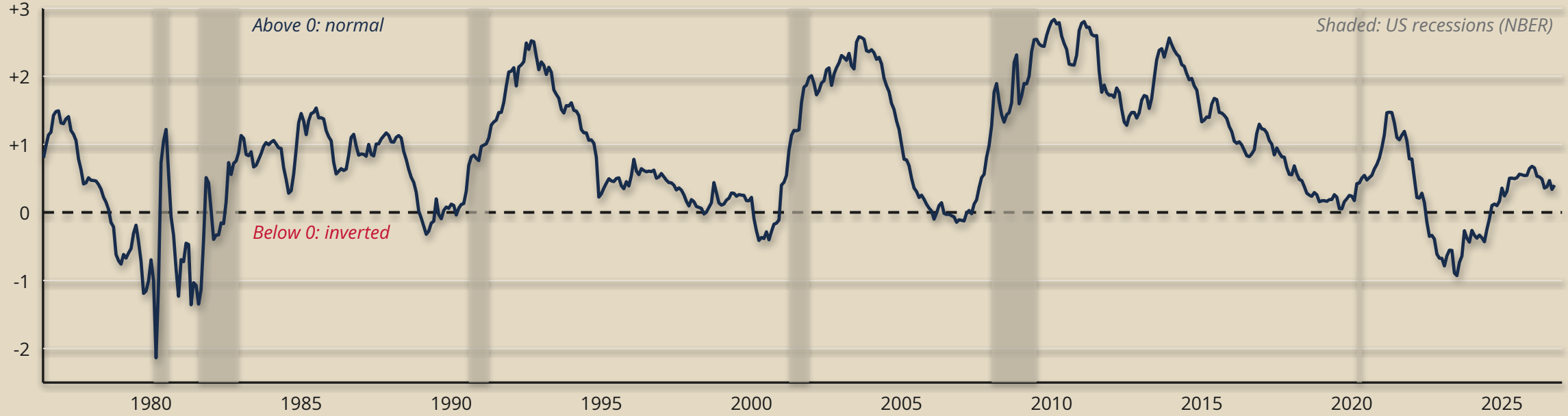
Warning sign
Markets expect Fed CUTS, often because a slowdown is coming → long < short.

Upward = short rates expected to rise or stay (plus a premium); inverted = cuts expected.

The Inverted Yield Curve: A Recession Signal?

Fifty years of US data

10-year minus 2-year Treasury yield (percentage points)



Source: FRED, T10Y2Y (monthly average of daily data, Jun 1976–Oct 2026) and USREC.

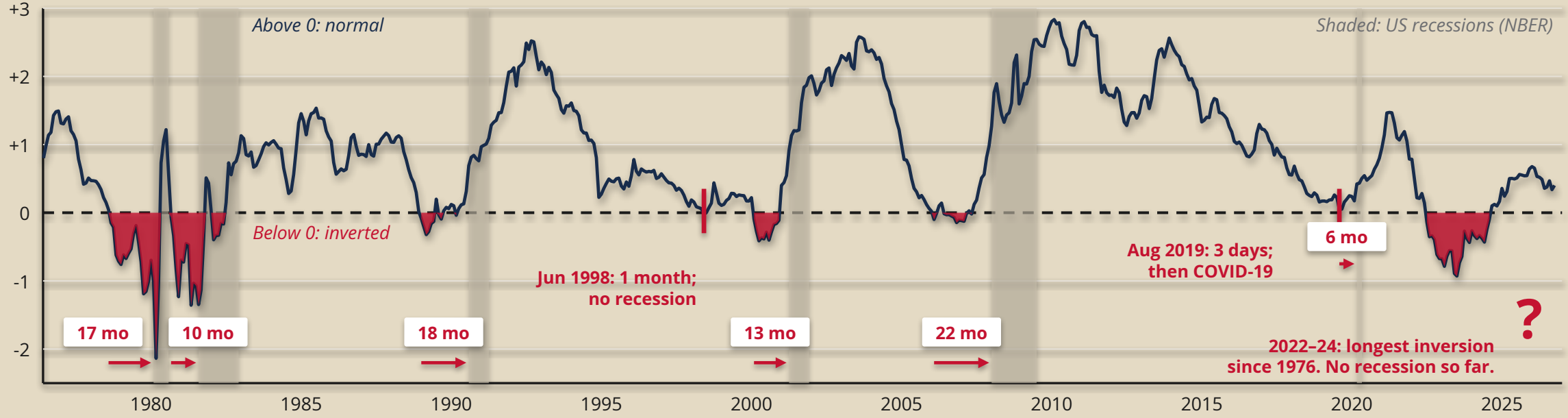
What do you see just BEFORE each shaded recession?

Look at where the line is relative to 0.

The Inverted Yield Curve: A Recession Signal?

Inversions came before every recession, but not every inversion brought one

10-year minus 2-year Treasury yield (percentage points)



Source: FRED, T10Y2Y (monthly average of daily data, Jun 1976–Oct 2026) and USREC.

Why it works: the logic chain

Markets expect a slowdown → expect the Fed to cut → long yields fall below short yields (slide 29: long ≈ expected short + premium).

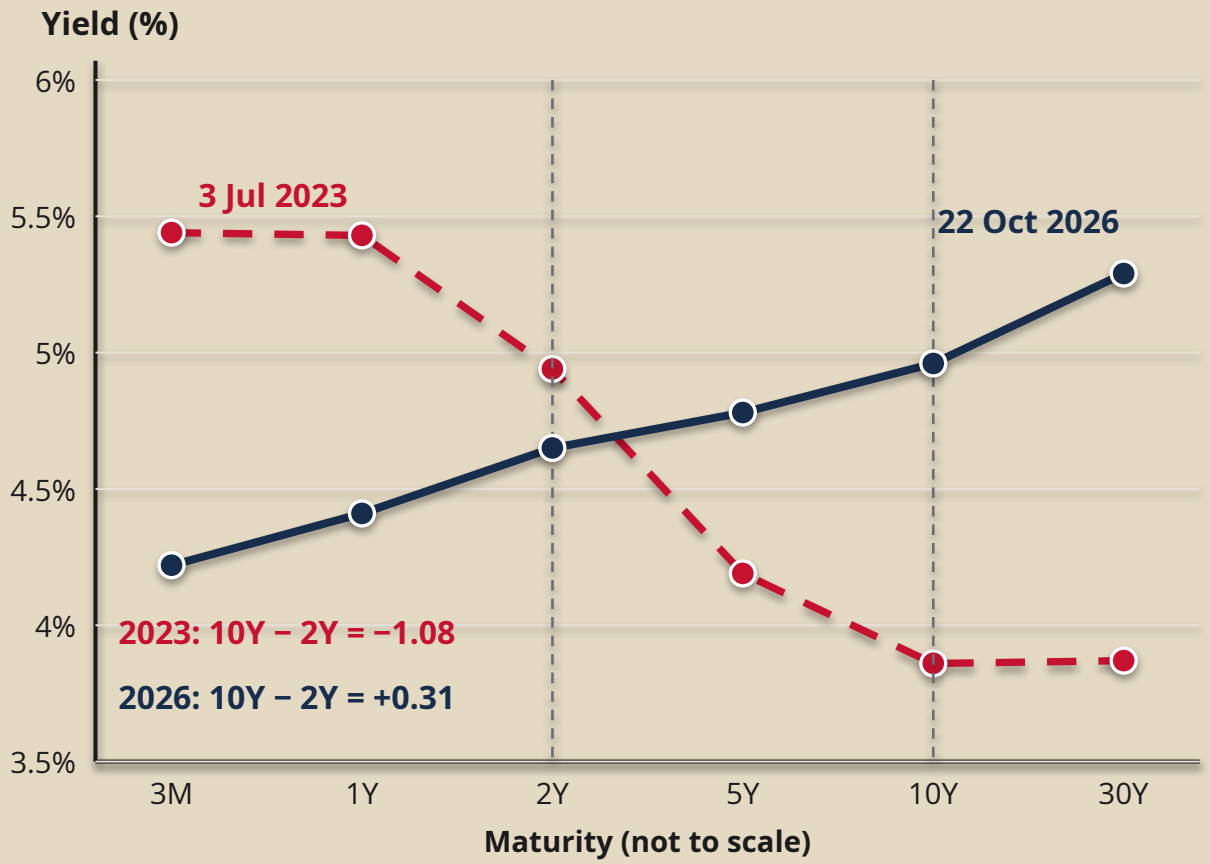
Track record since 1976

All 6 recessions came 6–22 months after an inversion. False alarms: a brief one in 1998, and 2022–24 so far (as of Oct 2026).

An inverted curve is a warning, not a guarantee.

Reading the Yield Curve: Then and Now

July 2023 vs. October 2026



Source: FRED, US Treasury constant-maturity yields. Axis starts at 3.5%.

Then: 3 July 2023

Fed target 5.00–5.25% after 10 hikes since March 2022. Markets expect cuts later → long < short.

Now: 22 October 2026

Fed target 3.75–4.00%: cuts in 2024–25, one hike in Sep 2026. Long > short again.

What changed?

Short end DOWN (3-month 5.44% → 4.22%): the Fed cut. Long end UP (10-year 3.86% → 4.96%): markets no longer expect big cuts ahead (slide 29).

Qatar: the riyal is pegged to the dollar (QR 3.64 = $1), so QCB moves its policy rates with the Fed. Qatar's short-term rates follow the US short end.

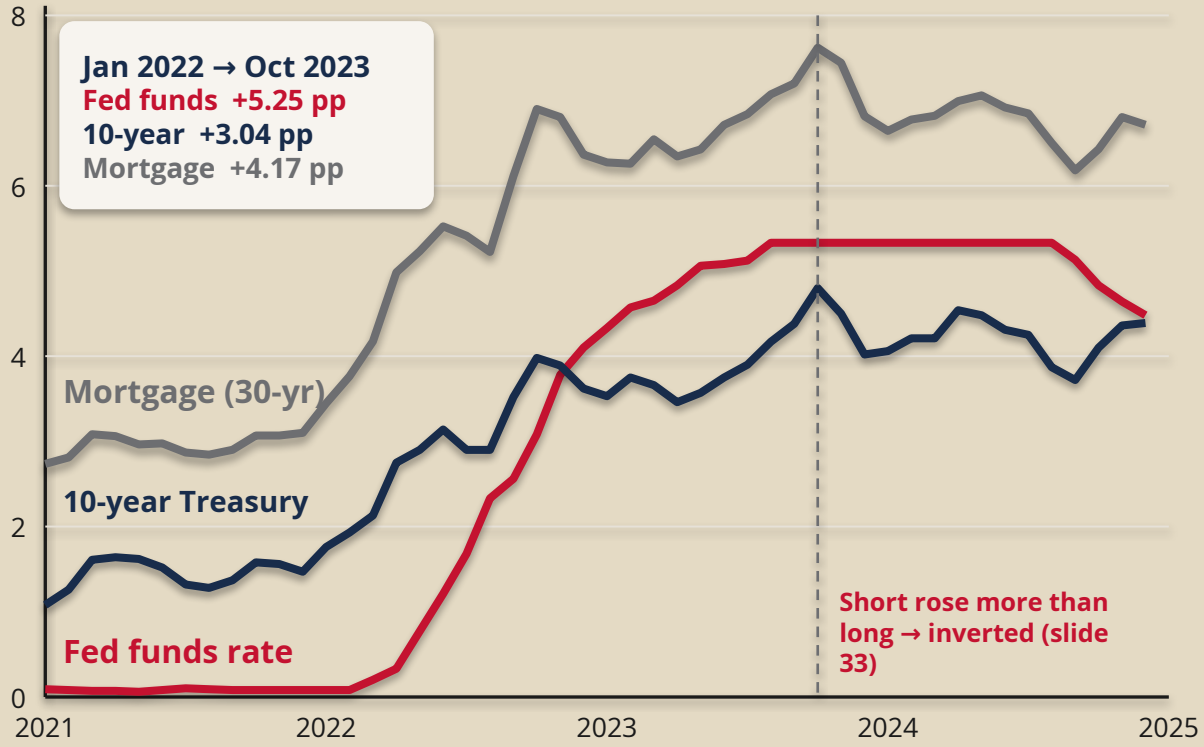
Discuss (2 min): the Fed's rate is LOWER than in 2023, yet the 10-year yield is HIGHER. Why?

The curve moves with expectations: inverted in 2023 (cuts expected), upward again in 2026.

Why the Yield Curve Matters for Macroeconomics

The bridge between central bank policy and the real economy

Interest rates (%), monthly, 2021–2024



Source: FRED (FEDFUNDS, GS10, MORTGAGE30US), monthly averages.

1. The Fed: open market operations set the short rate (slides 21, 25)

↓

2. Markets: expected path of future short rates

↓

3. Long rates = expected short rates + premium (slide 29)

↓

4. Borrowing costs: mortgages, car and business loans

↓

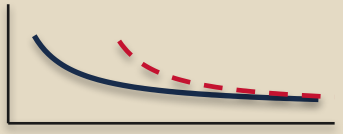
5. Spending: housing and investment → output and jobs

Coming up: Chapter 5 (IS–LM) uses a single interest rate, i. Chapter 14 shows why long rates are the ones that matter for spending.

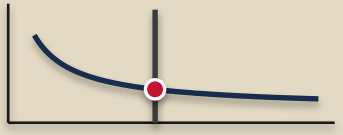
The Fed sets the short rate; through expectations, long rates and spending follow.

Session 1: Key Takeaways

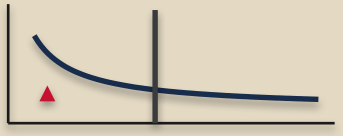
1. Money demand: $M^d = \$Y \cdot L(i)$. \$Y shifts the curve; a change in i is a move along it.



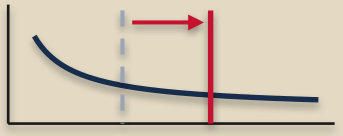
2. The central bank sets M^s; equilibrium $M^s = M^d$ determines i.



3. Out of equilibrium, people buy or sell bonds: $i = (\$100 - P_B) / P_B$ moves until $M^d = M^s$.



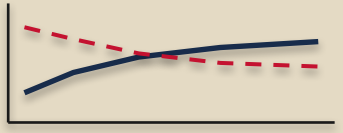
4. Open market operations: buy bonds → M^s ↑ → P_B ↑ → i ↓ (selling is the mirror image).



5. Bond prices and yields move in opposite directions: $P_B = \$100 / (1 + i)$. Long bonds swing more.



6. Yield curve: long rate ≈ expected short rates + premium. Inverted = cuts expected: a warning, not a guarantee.



One model: M^d, the central bank's M^s, and the bond market doing the adjusting.

Session 1 Recap & Preview

What we covered

- Money vs. bonds: liquidity vs. return
- Money demand: $M^d = \$Y \cdot L(i)$
- Equilibrium, and how the bond market adjusts
- Open market operations
- Bond prices and yields
- The yield curve as a recession signal

Coming next session

- Banks as financial intermediaries (Figure 4-6)
- Demand for reserves; the market for central bank money (Figure 4-7)
- The federal funds market
- The liquidity trap and the zero lower bound (Figure 4-8)
- Negative interest rates; the COVID-19 monetary response

Exit question
Nominal income $\$Y$ doubles and the central bank keeps M^s fixed. What happens to i? Draw it, and explain the bond-market adjustment.

Before next class: review $M^d = \$Y \cdot L(i)$, and think about who creates most of our money: banks.

Chapter 4 (continued) | Blanchard, Macroeconomics 9e

Today: What happens when we add banks to the model?

And what happens when interest rates hit zero?

Financial Markets I

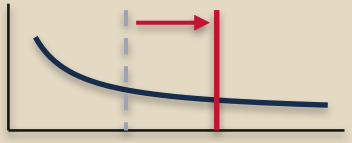
Session 2: Banks, Reserves & Liquidity Traps

Quick Review: Session 1

Let's make sure we remember the basics

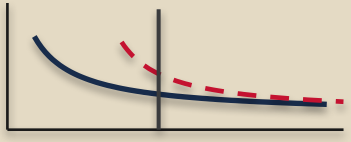
Q1
If the central bank increases M^s, what happens to i?

i falls
People buy bonds, P_B rises (slide 20)



Q2
If nominal income $\$Y$ rises, what happens to i?

i rises
M^d shifts right; people sell bonds (slide 18)



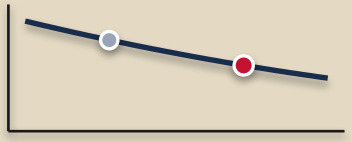
Q3
An expansionary open market operation means the central bank _____ bonds.

Buys
and pays with new money: M^s ↑ (slide 21)



Q4
If bond yields rise, what happens to bond prices?

They fall
$P_B = \$100 / (1 + i)$ (slides 26–28)



Today: add banks to this model (Section 4-3), then ask what happens when i hits 0 (Section 4-4).

SECTION 4-3

Determining the Interest Rate II

Adding banks to the model: financial intermediaries and reserves

- Banks as financial intermediaries
- Reserves and the monetary base
- The federal funds market

Banks as Financial Intermediaries

Figure 4-6: the balance sheets of the central bank and of a bank

Central bank

Assets			Liabilities		
Bonds	$100 →	**$150**	Reserves	$100 →	**$150**
			Currency (assume none)		**$0**

Its liabilities are central bank money, H = currency + reserves.

A commercial bank

Assets			Liabilities		
Reserves	$100 →	**$150**	Checkable deposits		**$1,000**
Loans		**$600**			
Bonds	$300 →	**$250**			

Takes deposits; lends, buys bonds, keeps some as reserves.

Same dollars: an asset for the bank, a liability for the central bank

Open market operation (slide 21): the central bank buys $50 of bonds from the bank
It pays by adding $50 to the bank's reserve account, so *H* rises by $50.

Suppose θ = 10%: the bank needs 10% × $1,000 = $100 of reserves. It now holds $150.
$50 more than it needs. What will it do with them? → the federal funds market (slide 45)

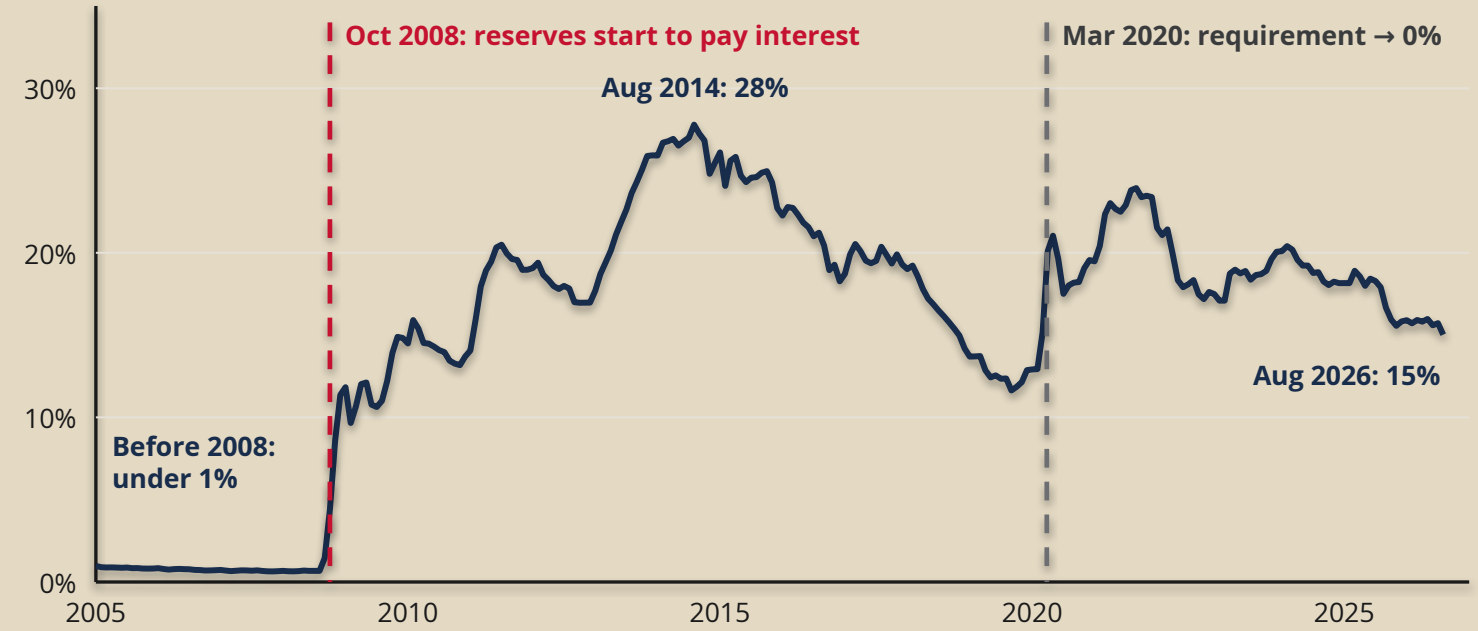
Central bank money *H* = reserves (+ currency). Open market operations change *H* one-for-one.

Why Do Banks Hold Reserves?

Reserves earn less than loans. So why hold them at all?

US banks must hold 0%. Why are reserves 15% of deposits?

US bank reserves as % of all bank deposits, 2005–Aug 2026



Source: FRED, TOTRESNS / DPSACBM027SBOG (commercial-bank deposits), monthly. As of Aug 2026.

1. Withdrawals

Some days customers take out more cash than they bring in. The bank must pay them on the spot.

2. Payments to other banks

When a customer pays someone at another bank, the banks settle in reserves. A bank that is short borrows overnight (slide 45).

3. The law may require it

US: 10% of checkable deposits until Mar 2020, 0% since. Qatar: 3.5% of deposits at QCB (4.5% until Mar 2026), earning nothing.

4. Since 2008, reserves pay interest

The Fed pays 3.90%, close to overnight lending (3.88%), so big reserves cost nothing. QE created them (slide 51).

Withdrawals, payments, rules and, since 2008, interest: θ (slide 43) sums up why banks hold reserves.

Demand for Central Bank Money

From people's demand for deposits to banks' demand for reserves (Blanchard eqs. 4.3–4.6)

Simplification (Blanchard 9e): people hold no currency, so all money is checkable deposits.

(4.3) $M^d = \$Y\, L(i)$

People's demand for deposits: same as Session 1

(4.4) $H^d = \theta\, M^d = \theta\, \$Y\, L(i)$

Banks keep a share θ of deposits as reserves

(4.5) $H = H^d$

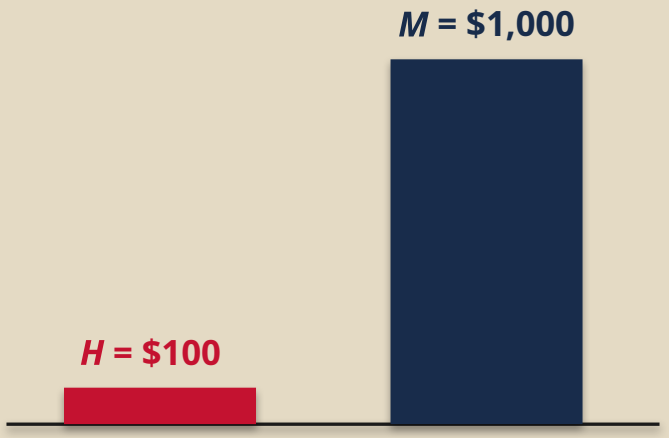
Supply of central bank money = demand

(4.6) $H = \theta\, \$Y\, L(i)$

Session 1 (slide 36): M = $Y L(i). Same equation, multiplied by θ, and now the market is for central bank money.

Example (slide 41)
θ = 10%, deposits $1,000 → reserves needed $100

Each $1 of *H* supports $1/*θ* of deposits

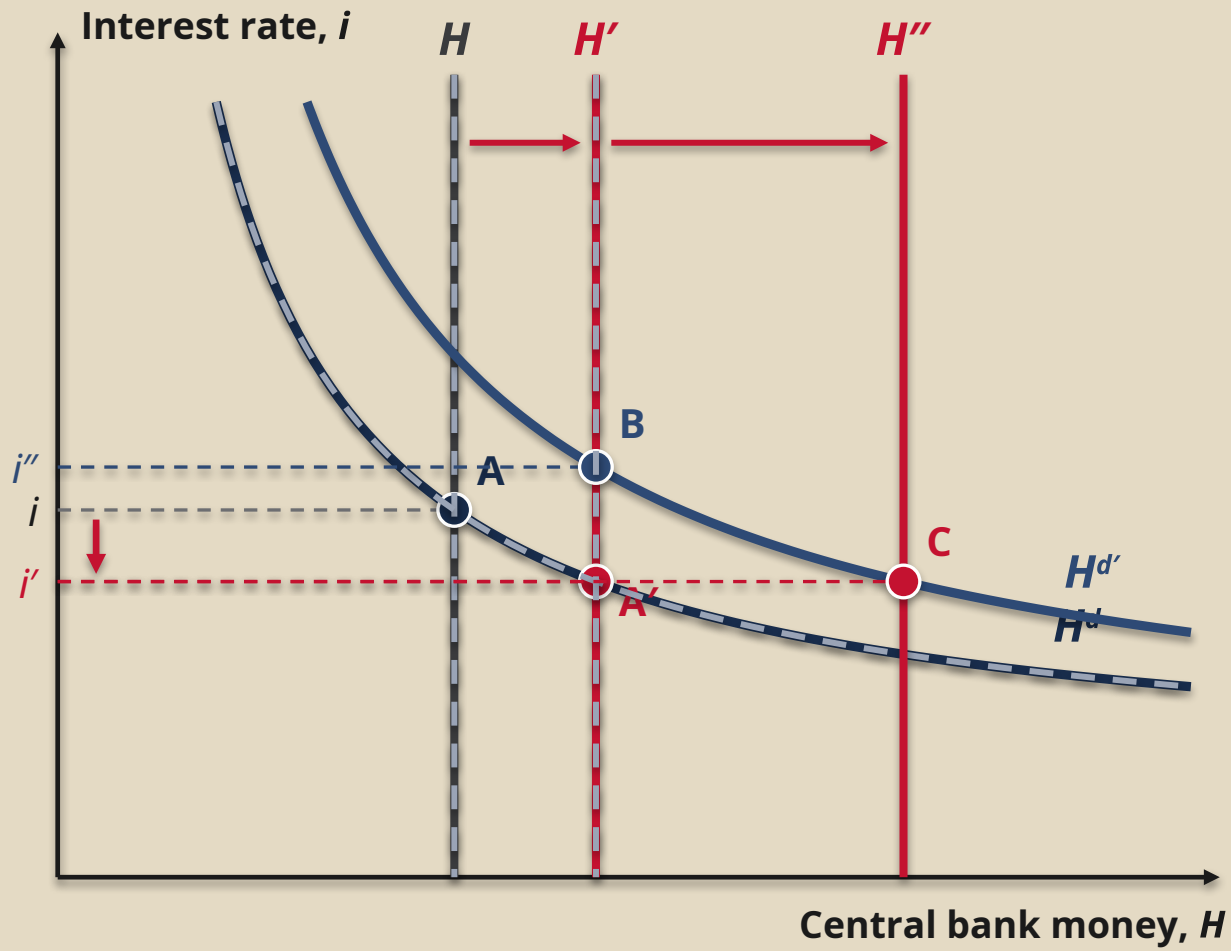


$M = H / \theta = \$100 / 0.10 = \$1{,}000$

Same logic as Session 1, but the market is now for central bank money: $H = \theta\, \$Y\, L(i)$.

Figure 4-7: The Market for Central Bank Money

Supply of central bank money meets banks' demand for reserves and determines i



Demand: $H^d = \theta \$Y\, L(i)$

Higher i → people want fewer deposits → banks need fewer reserves.

Supply: H, set by the central bank

A vertical line: it does not depend on i.

H ↑ → i ↓ (A to A′)

Banks hold more reserves than they need (slide 41), lend them overnight, and the rate falls until they are happy to hold all of H'.

$\$Y$ ↑ → H^d shifts right → i ↑ (B)

More deposits → banks need more reserves.

To keep i at i', the Fed adds H (C)

This is how it hits a target rate: it supplies whatever H banks demand at that rate (slide 45).

Same picture as slides 18–20, with *H* for *M*: by choosing *H*, the central bank chooses *i*.

The Federal Funds Market

Where banks trade reserves overnight: Figure 4-7 in real life

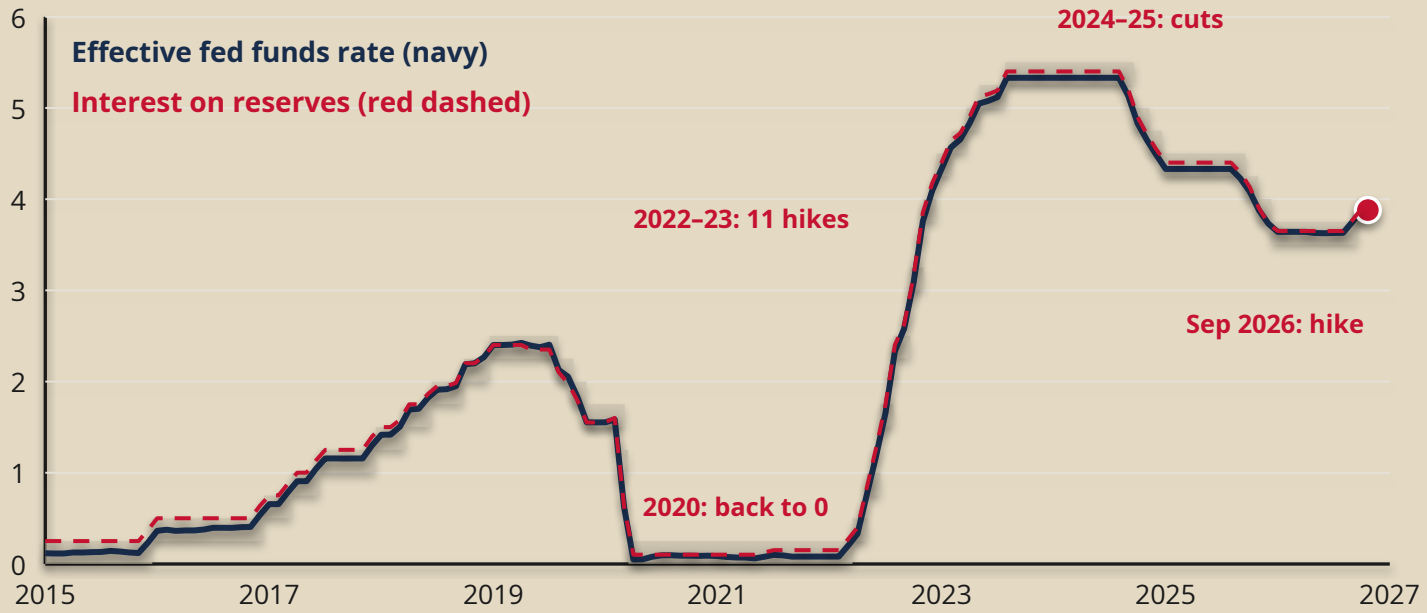
Bank A
has more reserves than it needs

lends reserves overnight →

← repays + interest at i_{ff}

Bank B
is short of reserves

Federal funds rate (%), 2015–Oct 2026



Source: FRED (DFEDTARL, DFEDTARU, EFFR, IOER, IORB). Shaded band = the Fed's target range.

As of 22 Oct 2026
Effective rate 3.88%; Fed target 3.75–4.00% (since 17 Sep 2026)

Qatar: QCB moved the same day
Peg: QR 3.64 = $1. On 16 Sep 2026 QCB raised its rates by 0.25 pp: deposit 4.10%, repo 4.35%, lending 4.60%. QIBOR, Qatar's interbank rate, moves with them.

Modern twist: interest on reserves
Banks now hold $2.96T of reserves (21 Oct 2026). The Fed steers i_{ff} mainly by setting the rate it pays on them (3.90%); i_{ff} trades just below it (3.88%).

All other rates are built on this one
It anchors the short end of the yield curve (slides 29–31).

The fed funds market is the market for central bank money: the Fed sets i_{ff}; Qatar's rates follow.

SECTION 4-4

The Liquidity Trap

When monetary policy becomes powerless: the zero lower bound

- Why interest rates can't go below zero (usually)
- The liquidity trap explained
- Real-world examples: 2008, COVID, negative rates

The Liquidity Trap

Why can't the central bank push interest rates far below zero?

Could the Fed push the bond rate to −2%?

From slide 26: $P_B = \$100 / (1 + i)$

$i = 5\% \rightarrow P_B = \$95.24$ $\quad$ $i = 0\% \rightarrow P_B = \$100$

At −2% you pay \$102.04 today to get \$100 next year.
Or you hold cash and keep your \$100. Nobody buys the bond, so i cannot go (much) below 0.

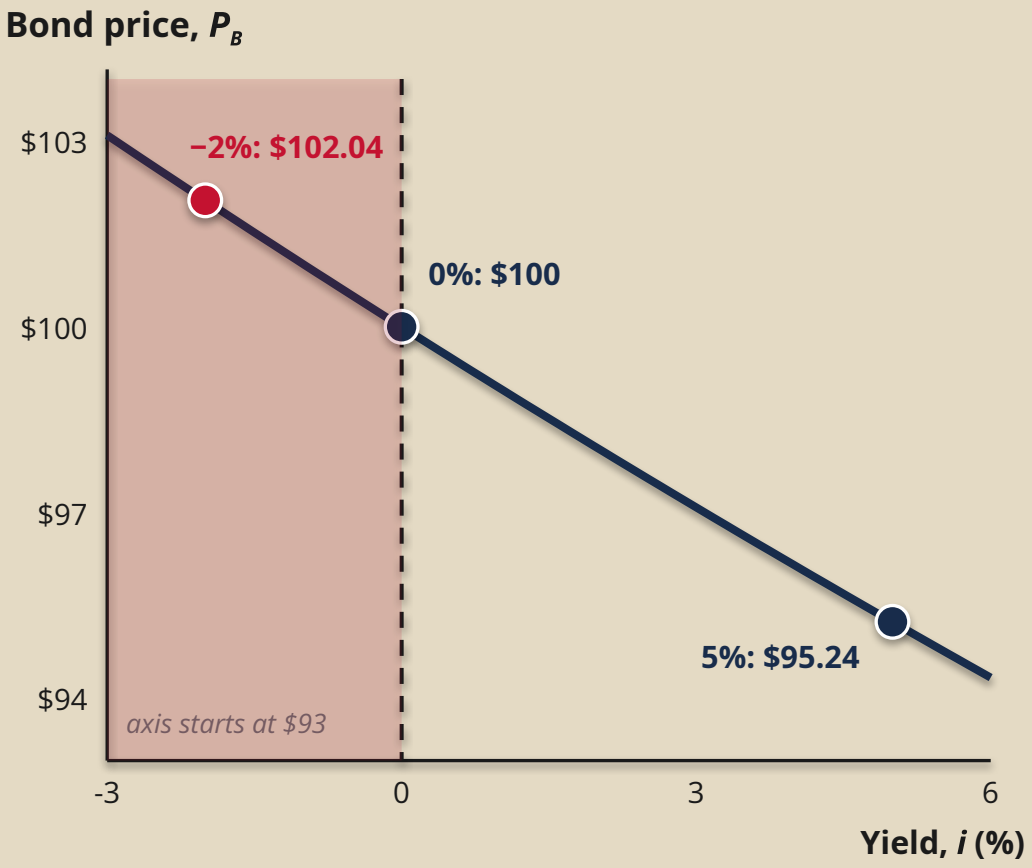
In practice rates can go slightly below 0: storing and moving piles of cash is costly (slide 52).

Normal times

$M^s \uparrow \rightarrow i \downarrow \rightarrow$ spending ↑

At the zero lower bound

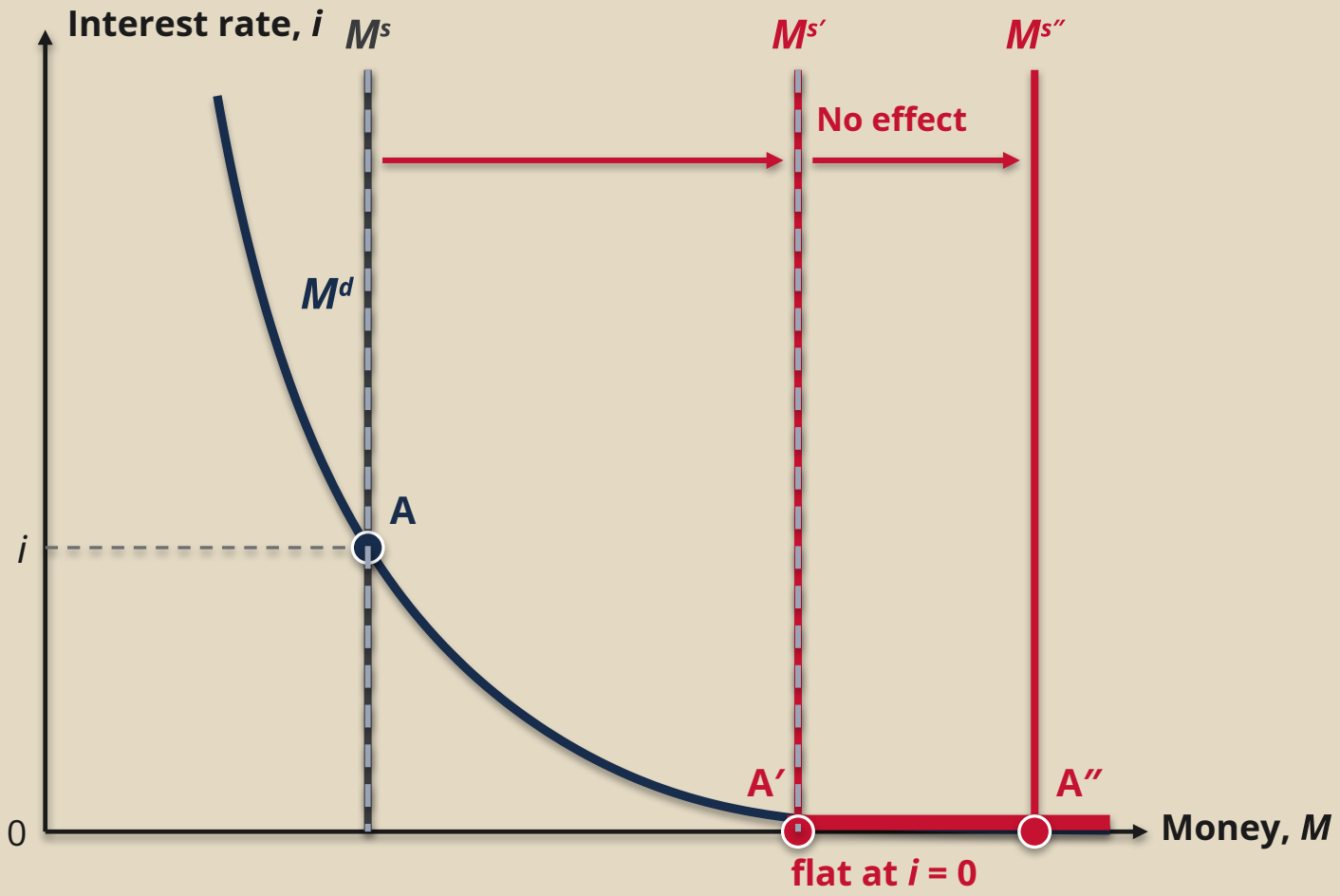
$M^s \uparrow \rightarrow i$ stays at 0



Money pays 0%, so bonds can't pay much less: the zero lower bound on i.

Figure 4-8: The Liquidity Trap

Money demand becomes flat at i = 0



Money demand, M^d

For $i > 0$: people hold more money only if i falls.

At $i = 0$, M^d is flat

Bonds pay nothing extra, so people are indifferent: they will hold any amount of money.

A → A′: i falls to 0

Normal case, as on slide 20: more money → people buy bonds → i ↓, until it reaches 0.

A′ → A″: i stays at 0

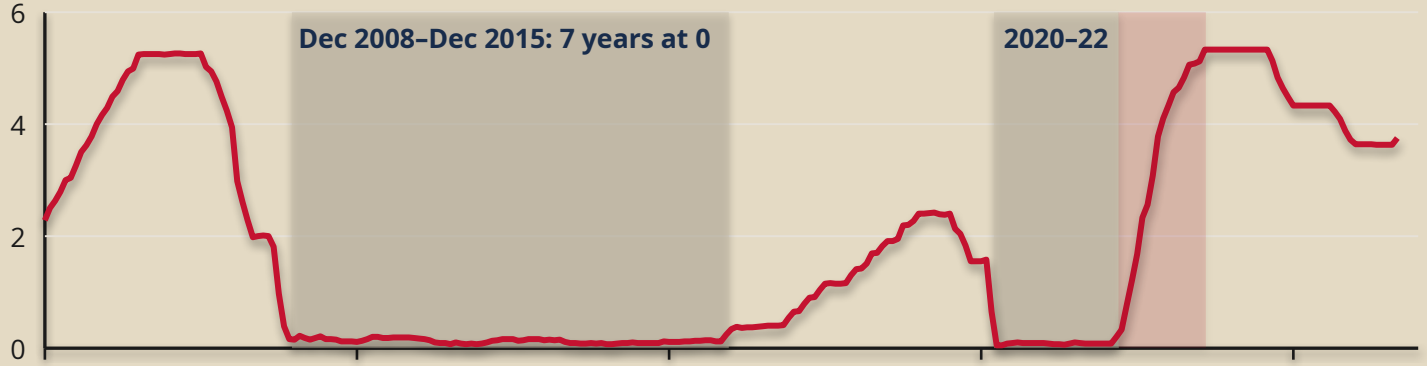
Already at the zero lower bound. The extra money is simply held: banks sit on reserves (slide 51).

In a liquidity trap, more money cannot lower i: monetary policy loses its usual power.

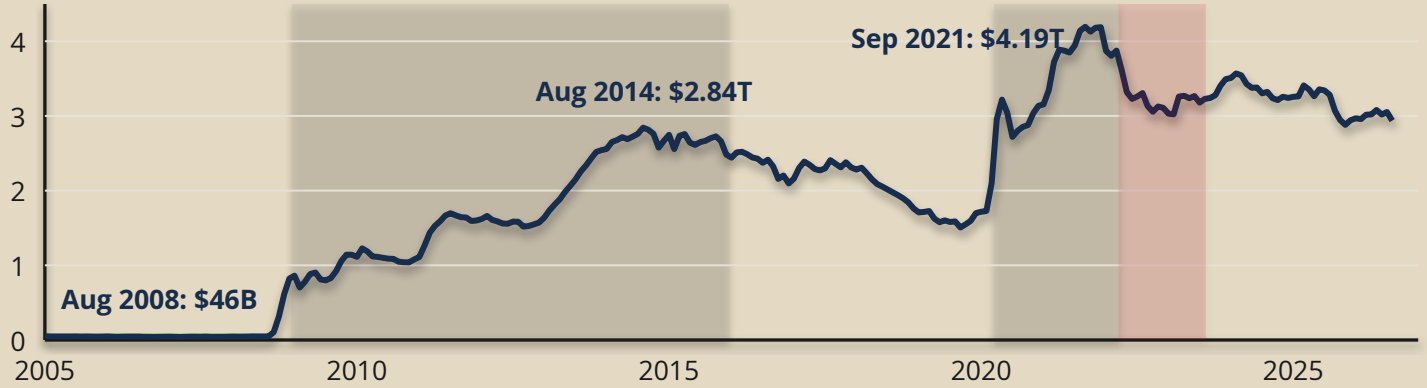
Focus: The Liquidity Trap in Action

US data, 2005 to Oct 2026 (FRED)

Federal funds rate (%)



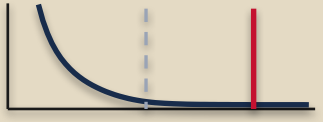
Bank reserves ($ trillion)



Source: FRED, FEDFUNDS and TOTRESNS (monthly), WRESBAL (weekly). As of Oct 2026.

The rate hit zero, twice
Fed target 0–0.25%: Dec 2008–Dec 2015 and Mar 2020–Mar 2022

At 0, more reserves did not lower *i*
Aug 2008: $46B, *i* = 2% (A). Dec 2008: $0.82T, *i* ≈ 0 (A′). Aug 2014: $2.84T, *i* still ≈ 0 (A″, slide 50).



Twist: 2022–23
The rate rose to 5.33% while reserves were still $3.0–3.9T. With so many reserves, the Fed moves *i* with the interest it pays on them (slide 45). Latest: $2.96T of reserves (21 Oct 2026), fed funds 3.88% (22 Oct).

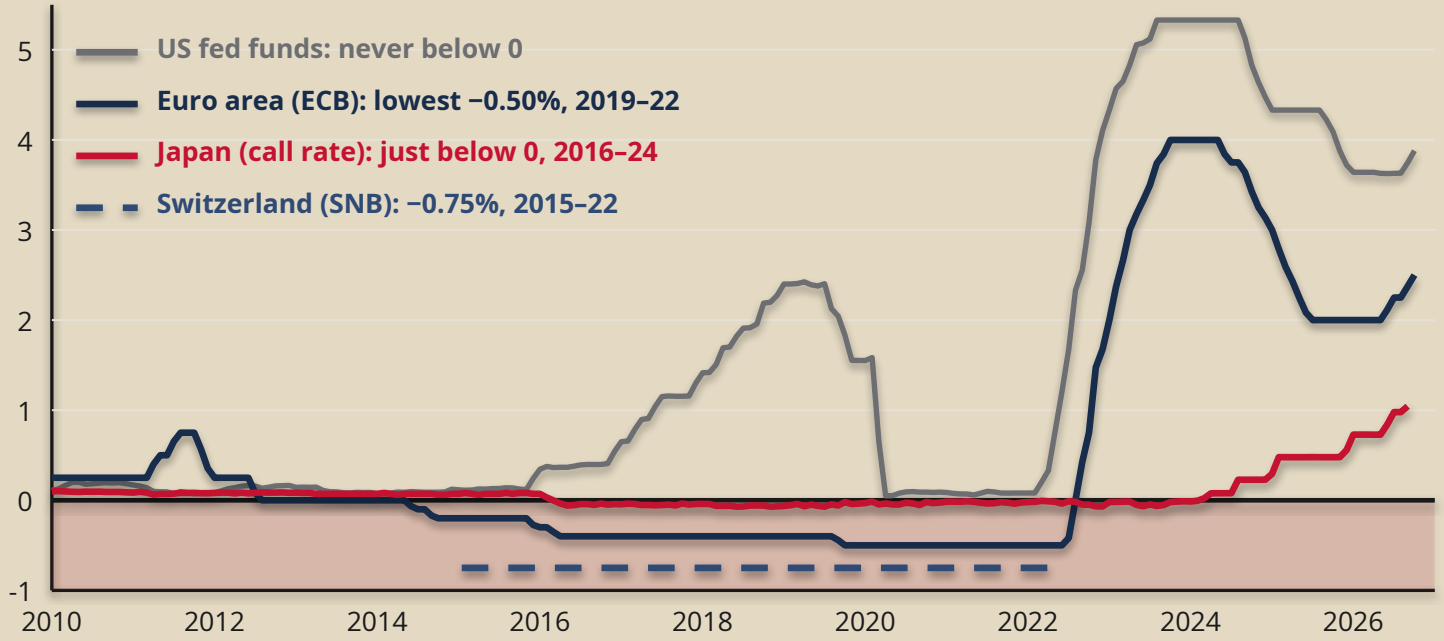
After 2008 the Fed flooded banks with reserves, yet *i* stayed at zero: the liquidity trap in action.

Modern Twist: Can Interest Rates Go Negative?

Some central banks tried. How far below zero did they go?

If cash pays 0%, can a central bank set negative interest rates?

Policy interest rates (%), 2010–Oct 2026



Source: FRED (DFF; ECBDFR; IRSTCI01JPM156N, Japan overnight call rate), monthly averages; SNB. As of Oct 2026.

Yes, but only a little

The lowest was −0.75%. Go much lower and people switch to cash, which pays 0% (slide 49).

Qatar: not an option

With the peg, QCB moves with the Fed, and the Fed never went below 0.

Other tools near zero

Quantitative easing (buy long bonds), forward guidance (promise low rates for long), fiscal policy. Next: COVID-19.

The floor is near zero, not exactly zero: negative rates exist, but only slightly below 0.

Case Study: COVID-19 Monetary Response

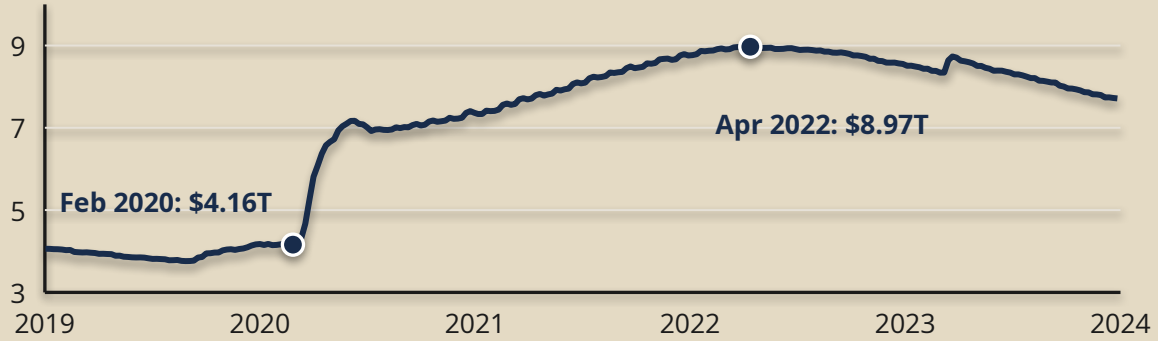
CASE STUDY

March 2020: lockdowns, and spending collapses. How did the Fed and QCB respond?

US federal funds rate (%), weekly



Fed balance sheet ($ trillion), weekly



Source: FRED, DFF and WALCL; QCB press releases (16 Mar 2020); Peninsula/QNA.

Qatar (QCB), March 2020

Two cuts, following the Fed: deposit rate to 1.00%, lending rate to 2.50%. A zero-interest repo window to free about QAR 50B for banks; six-month loan deferrals. The government added a QAR 75B package for the private sector.

Pairs, 3 minutes. Use Figure 4-8 (slide 50):

Q1: Was the US in a liquidity trap?

Q2: Was Qatar trapped? Why or why not?

Hint: compare where each policy rate ended up.

Case Study Solution: COVID-19

Two economies, two different limits on monetary policy

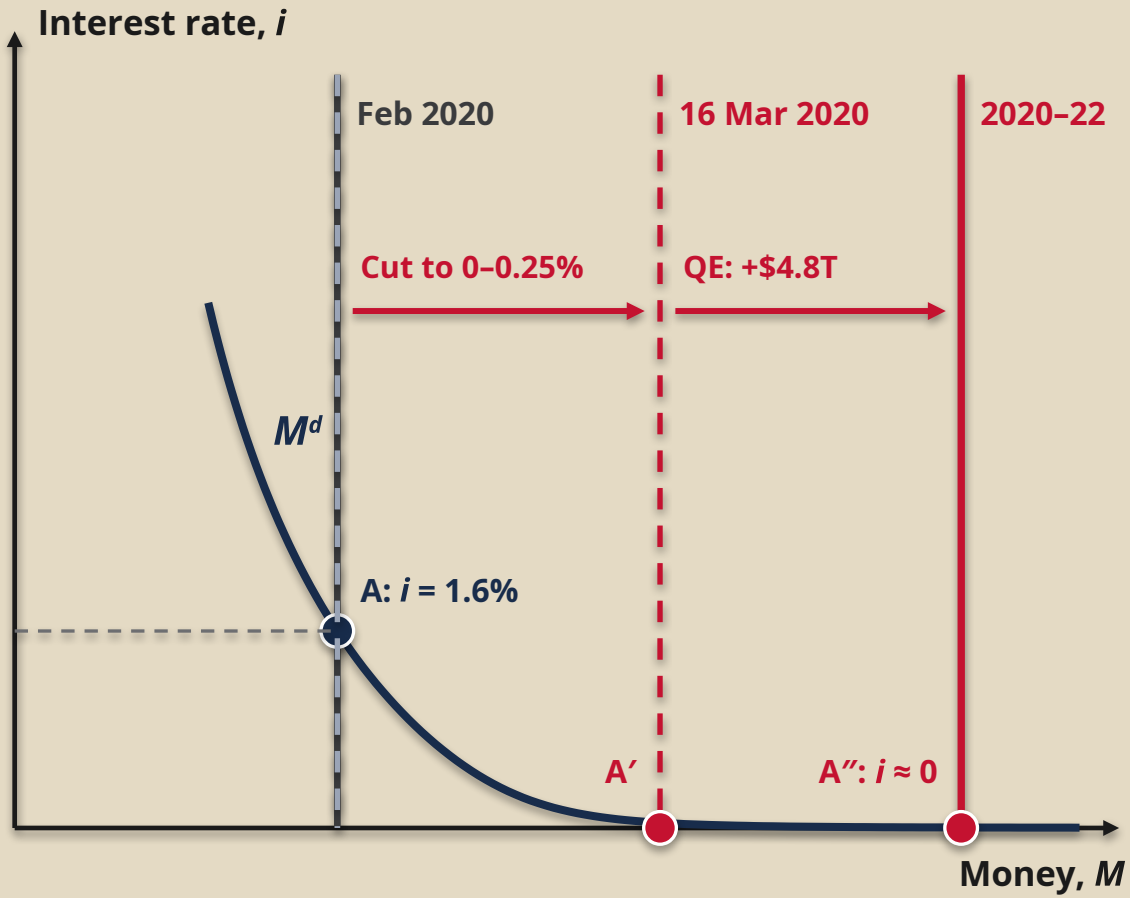


Diagram is schematic (Figure 4-8). Rates: FRED DFF; QCB press releases.

Q1. United States: yes

The rate hit 0 within days, then the Fed added $4.8T and *i* stayed at about 0 (point A″). So it used other tools: QE to push down long rates, forward guidance, and Congress passed the $2.2T CARES Act.

Q2. Qatar: not at the zero lower bound

QCB's deposit rate stayed at 1.00% from March 2020 into 2022: above 0. Its limit was the peg (QR 3.64 = $1): QCB follows the Fed instead of choosing its own rate. Fiscal policy (QAR 75B) did much of the work.

US limit
the zero lower bound

Qatar limit
the currency peg

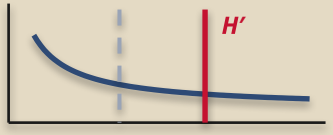
Two different limits on monetary policy: the zero lower bound (US) and the currency peg (Qatar).

Final Concept Check

Test your understanding of the complete model

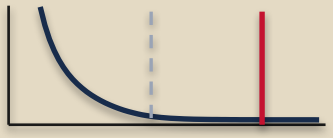
Q1
If the central bank increases H, what happens to i?

i falls: H shifts right along H^d (slide 44). Unless i is already 0 (slide 50)



Q2
In a liquidity trap, does increasing M^s lower i?

No: M^d is flat at $i = 0$, so i stays at 0 (slide 50)



Q3
What tools remain in a liquidity trap?

QE, forward guidance, and fiscal policy (slide 52)

Q4
Why does Qatar's QCB follow the Fed's rate decisions?

The riyal is pegged (QR 3.64 = $1), so Qatar's rates must track US rates (slides 45, 54)

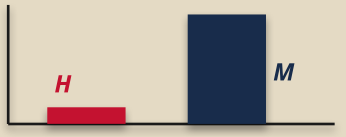
Q5
Why did bank reserves explode after 2008 while lending did not?

At $i \approx 0$, holding reserves cost banks nothing (and they earn interest on them); loan demand was weak. The extra H just sat (slides 50–51)

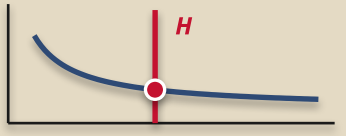
The complete model: $H = \theta \$Y L(i)$, the fed funds market, and the zero lower bound.

Session 2: Key Takeaways

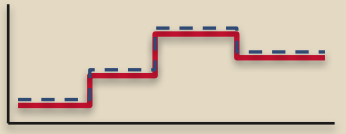
1. Banks keep a share θ of deposits as reserves. With no currency, $H = \theta M$, so $M = H / \theta$.



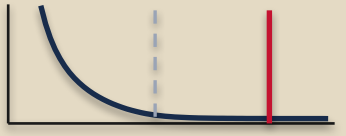
2. Equilibrium in the market for central bank money: $H = \theta \$Y L(i)$ (Figure 4-7).



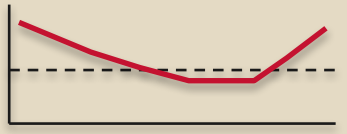
3. The federal funds rate is the price of reserves and the Fed's policy rate; QCB's rates track it (peg).



4. Liquidity trap: at $i = 0$, money demand is flat, so more money does not lower i (Figure 4-8).



5. Near zero: QE, forward guidance, fiscal policy. Negative rates go only slightly below 0.



Banks, reserves, and the zero lower bound complete the Chapter 4 model.

Chapter 4: Financial Markets I — Complete Summary

Model	Key equation	Policy tool	Limitation
Simple (4-2)	$M = \$Y\, L(i)$	Open market operations change M^s	Ignores banks
With banks (4-3)	$H = \theta\, \$Y\, L(i)$	Change H to hit the fed funds target	Assumes i can always fall
Liquidity trap (4-4)	At $i = 0$, M^d is flat	QE, forward guidance, fiscal policy	Effects harder to predict

- The central bank determines the short-term interest rate.
- Money demand depends on $\$Y$ and i.
- Bond prices and yields move inversely; the yield curve reflects expected future rates.
- With banks, \$1 of central bank money supports $\$1/\theta$ of deposits ($M = H / \theta$).
- At the zero lower bound, conventional monetary policy stops working.
- Next chapter: combine financial markets with the goods market (IS–LM).

Chapter 4 in one line: the central bank sets *i*, until *i* hits zero.